S73-M-Q1 · question page 1514

AI.IMAGE.02 · Inspect a generated image for visual errors and garbled text

Which part of a generated image often has errors?

Answer: D. Hands and fingers

Extra or missing fingers.

- ✗ A. The image size: not a visual flaw
- ✗ B. The overall color of the sky, which image tools always get wrong in every picture: not the common issue
- ✗ C. The file name: not part of the image

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-M-Q2 · question page 1515

AI.IMAGE.02 · Inspect a generated image for visual errors and garbled text

In this prepared example, which word is misspelled?

Prepared example (a written description of a fictional generated image)

Image description: a cheerful flyer with cupcakes. Visible text reads "SPRING BAKE SAEL - Saturday, April 31 - 10 am to 2 pm - Room 12."

Answer: C. SAEL

It should read SALE.

- ✗ A. Room: spelled correctly
- ✗ B. Saturday: spelled correctly
- ✗ D. SPRING: spelled correctly

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-M-Q3 · question page 1516

AI.IMAGE.02 · Inspect a generated image for visual errors and garbled text

In the same example, what other error appears?

Prepared example (a written description of a fictional generated image)

Image description: a cheerful flyer with cupcakes. Visible text reads "SPRING BAKE SAEL - Saturday, April 31 - 10 am to 2 pm - Room 12."

Answer: B. April 31 does not exist

April has 30 days.

- ✗ A. Room 12 is not a real number: it is fine
- ✗ C. Saturday is misspelled: it is correct
- ✗ D. 10 am to 2 pm is too long: not an error

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-M-Q4 · question page 1517

AI.IMAGE.02 · Inspect a generated image for visual errors and garbled text

How should you check text in an image?

Answer: A. Zoom in and read it letter by letter

Careful reading.

- ✗ B. Trust it if the image is pretty: not a check
- ✗ C. Count the colors: not about text
- ✗ D. Glance at it quickly from a distance: misses errors

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-M-Q5 · question page 1518

AI.IMAGE.02 · Inspect a generated image for visual errors and garbled text

What is often easier than fixing text the generator made?

Answer: B. Adding the text yourself in a design app

Add text yourself.

- ✗ A. Leaving the errors in: not good
- ✗ C. Removing all images: unnecessary
- ✗ D. Asking for the same image again and again: unreliable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-B1-Q1 · question page 1519

AI.IMAGE.02 · Inspect an image with a checklist

Six fingers on a waving hand fails which point?

Answer: D. Bodies

Count fingers.

- ✗ A. Text: not text
- ✗ B. Patterns: not patterns
- ✗ C. Light: not light

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-B1-Q2 · question page 1520

AI.IMAGE.02 · Inspect an image with a checklist

A cup merging into a table fails which point?

Answer: A. Objects

Blending objects.

- ✗ B. Text: not text
- ✗ C. Light: not light
- ✗ D. Bodies: not stated

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-B1-Q3 · question page 1521

AI.IMAGE.02 · Inspect an image with a checklist

A sign reading "Opne" fails which point?

Answer: C. Text

Misspelled word.

- ✗ A. Objects: not objects
- ✗ B. Bodies: not bodies
- ✗ D. Patterns: not patterns

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-B1-Q4 · question page 1522

AI.IMAGE.02 · Inspect an image with a checklist

Shadows falling in two different directions fail which point?

Answer: D. Light

Inconsistent lighting.

- ✗ A. Text: no
- ✗ B. Bodies: no
- ✗ C. Objects: not the category

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-B1-Q5 · question page 1523

AI.IMAGE.02 · Inspect an image with a checklist

A striped shirt whose stripes change direction fails which point?

Answer: C. Patterns

Broken pattern.

- ✗ A. Bodies: not bodies
- ✗ B. Light: not light
- ✗ D. Text: not text

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-F1-Q1 · question page 1524

AI.IMAGE.02 · Catch image flaws before using an image

A poster image shows the event phone number with one digit changed. What should you do?

Answer: B. Fix the number before printing

Wrong details cause problems.

- ✗ A. Remove the event date too: unrelated
- ✗ C. Make the number smaller: still wrong
- ✗ D. Print it anyway: wrong contact

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-F1-Q2 · question page 1525

AI.IMAGE.02 · Catch image flaws before using an image

A family portrait for a slideshow has a child with three arms. What is the best choice?

Answer: A. Regenerate, edit or choose another image

Do not use flawed images.

- ✗ B. Make it black and white: still wrong
- ✗ C. Keep it, since viewers rarely look closely at people in pictures shown during slideshows: noticeable
- ✗ D. Add a caption explaining it: still odd

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-F1-Q3 · question page 1526

AI.IMAGE.02 · Catch image flaws before using an image

A store sign in a generated image reads "Grocey Stroe." What kind of flaw is this?

Answer: B. Garbled text

Misspelled text.

- ✗ A. No flaw: misspelled
- ✗ C. A pattern flaw: not pattern
- ✗ D. A lighting flaw: it is spelling

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-F1-Q4 · question page 1527

AI.IMAGE.02 · Catch image flaws before using an image

Before posting a generated image online, what is the best first step?

Answer: C. Zoom in and check details

Inspect first.

- ✗ A. Post it right away, since generated images are checked for errors before you receive them: not guaranteed
- ✗ B. Add a filter: not a check
- ✗ D. Compress it: not a check

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-F1-Q5 · question page 1528

AI.IMAGE.02 · Catch image flaws before using an image

A flyer has correct text that you added yourself and a generated background. What still needs checking?

Answer: D. The background for visual flaws

Check all generated parts.

- ✗ A. Only the font: not enough
- ✗ B. Nothing, because adding text yourself also fixes every flaw in the generated background: not true
- ✗ C. Only the file size: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-F2-Q1 · question page 1529

AI.IMAGE.02 · Prioritize the flaws that matter

Which flaw matters most on an event flyer?

Answer: A. A wrong date

Wrong facts cause real problems.

- ✗ B. A tree with odd leaves: minor
- ✗ C. A slightly blurry corner: minor
- ✗ D. A cloud in the background: harmless

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-F2-Q2 · question page 1530

AI.IMAGE.02 · Prioritize the flaws that matter

A certificate image misspells the winner's name. Why is this serious?

Answer: D. It gives wrong information about a real person

Names must be right.

- ✗ A. It changes the paper size: irrelevant
- ✗ B. It is not serious, because names on certificates are decorative and nobody reads them: they matter
- ✗ C. It makes the file larger: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-F2-Q3 · question page 1531

AI.IMAGE.02 · Prioritize the flaws that matter

A generated safety sign looks almost like a real one but the symbol is slightly different. What is the risk?

Answer: B. People may misread the warning

Wrong symbols mislead.

- ✗ A. The colors fade: not the risk
- ✗ C. No risk, because people always understand safety signs even when the symbol is changed: not guaranteed
- ✗ D. The sign becomes funny: not the risk

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-F2-Q4 · question page 1532

AI.IMAGE.02 · Prioritize the flaws that matter

An image looks fine on a phone. Why check before printing a large banner?

Answer: A. Flaws can be obvious when enlarged

Size reveals details.

- ✗ B. Because banners never show text: they do
- ✗ C. Because printing always changes every color in a picture into completely different colors: not the point
- ✗ D. Because phones show extra details: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-F2-Q5 · question page 1533

AI.IMAGE.02 · Prioritize the flaws that matter

A generated logo looks like another company's logo. What should you do?

Answer: C. Avoid using it

It may belong to someone else.

- ✗ A. Add your name and use it: still similar
- ✗ B. Change one color and use it: still similar
- ✗ D. Use it: they can

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-M-Q1 · question page 1535

AI.IMAGE.03 · Keep a series consistent and request targeted edits

What is drift in an image series?

Answer: C. Details changing from one image to the next

Changing hair, clothes or style.

- ✗ A. Images loading slowly: not the meaning
- ✗ B. Images getting larger: not the meaning
- ✗ D. Images slowly moving across the screen: not the meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-M-Q2 · question page 1536

AI.IMAGE.03 · Keep a series consistent and request targeted edits

How can you keep a character consistent?

Answer: A. Reuse the same detailed description each time

Same words, same features.

- ✗ B. Change the style each page: causes drift
- ✗ C. Leave out the description: causes drift
- ✗ D. Describe the character differently each time: causes drift

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-M-Q3 · question page 1537

AI.IMAGE.03 · Keep a series consistent and request targeted edits

Which edit request is targeted?

Answer: D. "Make only the raincoat green; keep everything else."

One clear change.

- ✗ A. "Make it different.": vague
- ✗ B. "Change it.": vague
- ✗ C. "Redraw the whole picture again from the beginning.": not targeted

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-M-Q4 · question page 1538

AI.IMAGE.03 · Keep a series consistent and request targeted edits

After an edit, what should you do?

Answer: B. Compare the result with the original

Compare versions.

- ✗ A. Post it immediately: not checked
- ✗ C. Make five more edits without looking: risky
- ✗ D. Delete the original image right away: loses comparison

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-M-Q5 · question page 1539

AI.IMAGE.03 · Keep a series consistent and request targeted edits

What tool feature can help consistency?

Answer: A. Reference images

Show the tool what to match.

- ✗ B. A faster internet connection: irrelevant
- ✗ C. A larger screen: not a tool feature
- ✗ D. A feature that makes every image: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-B1-Q1 · question page 1540

AI.IMAGE.03 · Compare a series with a school uniform

What does a uniform keep the same?

Answer: D. Colors, clothes and logo

Consistent features.

- ✗ A. The timetable: no
- ✗ B. The weather: no
- ✗ C. Each student's height: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-B1-Q2 · question page 1541

AI.IMAGE.03 · Compare a series with a school uniform

Students choosing new outfits daily is like what?

Answer: C. Drift in an image series

Changing details.

- ✗ A. A targeted edit: no
- ✗ B. A reference image: no
- ✗ D. A perfectly consistent series: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-B1-Q3 · question page 1542

AI.IMAGE.03 · Compare a series with a school uniform

A character's "uniform" is what?

Answer: B. The same written description used every time

Reuse the description.

- ✗ A. A new style each page: drift
- ✗ C. No description: drift
- ✗ D. A different description for every page: causes drift

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-B1-Q4 · question page 1543

AI.IMAGE.03 · Compare a series with a school uniform

Why do mismatched photos look odd?

Answer: A. Things that should match do not

Inconsistency is noticeable.

- ✗ B. Because uniforms are blue: irrelevant
- ✗ C. Because photos taken on different days always come out blurry and unusable: not the point
- ✗ D. Because schools are large: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-B1-Q5 · question page 1544

AI.IMAGE.03 · Compare a series with a school uniform

What is the lesson for an image series?

Answer: D. Keep the same details each time

Consistency.

- ✗ A. Use only one image: not the lesson
- ✗ B. Avoid characters: not the lesson
- ✗ C. Change everything each time: confusing

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-F1-Q1 · question page 1545

AI.IMAGE.03 · Keep series consistent and edit precisely

A children's story has a dog that is brown on page 1 and gray on page 3. What should the author do?

Answer: C. Reuse one exact description of the dog for every page

A fixed description.

- ✗ A. Use gray on every page without saying so: still drift
- ✗ B. Remove the dog: unnecessary
- ✗ D. Change the story so that the dog changes color by magic: unnecessary

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-F1-Q2 · question page 1546

AI.IMAGE.03 · Keep series consistent and edit precisely

Nia wants to change only the background of an image from a park to a beach. Which request fits?

Answer: B. "Change only the background to a beach; keep the people the same."

Targeted edit.

- ✗ A. "Beach.": vague
- ✗ C. "Change everything to look like summer.": too broad
- ✗ D. "Make a new beach picture with some people in it who look somewhat like the old ones.": not targeted

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-F1-Q3 · question page 1547

AI.IMAGE.03 · Keep series consistent and edit precisely

A slide series uses watercolor on some slides and photos on others. What should change?

Answer: A. Use the same style words on every slide

Consistent style words.

- ✗ B. Use random styles: less consistent
- ✗ C. Add more styles, so the slides look varied and keep the audience's attention throughout: less consistent
- ✗ D. Remove all images: unnecessary

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-F1-Q4 · question page 1548

AI.IMAGE.03 · Keep series consistent and edit precisely

After asking to change a shirt to blue, what should you check?

Answer: C. That only the shirt changed

Compare with the original.

- ✗ A. Only the file name: not enough
- ✗ B. Nothing, because a targeted edit request always changes exactly one thing in any picture: not guaranteed
- ✗ D. Only the size: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-F1-Q5 · question page 1549

AI.IMAGE.03 · Keep series consistent and edit precisely

Which description helps a character stay consistent?

Answer: B. "Leo: tall, red cap, green scarf, freckles."

Specific, reusable features.

- ✗ A. "A boy.": too vague
- ✗ C. "Leo.": too vague
- ✗ D. "Leo: a boy, and he looks however you think is best for the story on each page.": invites drift

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-F2-Q1 · question page 1550

AI.IMAGE.03 · Catch edits that change too much

You asked to change the sky, and the person's face also changed. What does this show?

Answer: D. Edits can change more than asked

Check the whole image.

- ✗ A. That the sky and face are the same: no
- ✗ B. That edits never work: too strong
- ✗ C. That the tool always changes faces: not on purpose

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-F2-Q2 · question page 1551

AI.IMAGE.03 · Catch edits that change too much

Why keep a copy of each version during edits?

Answer: A. So you can go back if an edit goes wrong

Undo.

- ✗ B. So images look brighter: not the reason
- ✗ C. To use more storage: not the reason
- ✗ D. Because tools refuse to edit any image: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-F2-Q3 · question page 1552

AI.IMAGE.03 · Catch edits that change too much

What is a style sheet for an image series?

Answer: B. A list of each character's features, colors and style

A reference for consistency.

- ✗ A. A list of fonts only: too narrow
- ✗ C. A spreadsheet of prices: no
- ✗ D. A document that lists the printer settings for every page: not the meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-F2-Q4 · question page 1553

AI.IMAGE.03 · Catch edits that change too much

A new image almost matches, but the character's glasses are missing. What should you do?

Answer: C. Reject or fix it using the style sheet

Check against the style sheet.

- ✗ A. Remove glasses from all other images: more work
- ✗ B. Change the character's name: irrelevant
- ✗ D. Accept it: readers notice

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-F2-Q5 · question page 1554

AI.IMAGE.03 · Catch edits that change too much

When should you check a new image against the style sheet?

Answer: D. Before accepting it

Check each image.

- ✗ A. After the whole series is printed and handed out, so you can fix it next time: too late
- ✗ B. Only for the first image: not enough
- ✗ C. Never: loses consistency

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-M-Q1 · question page 1556

AI.IMAGE.04 · Check generated diagrams, maps and charts against real data

Do image generators check facts when they draw a map?

Answer: A. No; they make pictures that look like maps

Look, not accuracy.

- ✗ B. Yes; every generated map is checked against official satellite data before it is shown: not true
- ✗ C. Only for small countries: no
- ✗ D. Only in color: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-M-Q2 · question page 1557

AI.IMAGE.04 · Check generated diagrams, maps and charts against real data

In this prepared example, which bar is drawn wrongly?

Prepared example (a written description of a fictional generated bar chart)

Bar label	Value printed on bar	Bar height drawn (grid squares)
Monday	10	1
Tuesday	40	4
Wednesday	20	5

Answer: B. Wednesday

At one square per 10, Wednesday (20) should be 2 squares, not 5.

- ✗ A. Tuesday: 40 drawn as 4 squares matches
- ✗ C. None: heights do not match
- ✗ D. Monday: 10 as 1 square matches

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-M-Q3 · question page 1558

AI.IMAGE.04 · Check generated diagrams, maps and charts against real data

How should you make a chart meant to inform?

Answer: C. Build it from real data, such as in a spreadsheet

Use real data.

- ✗ A. Copy a chart from memory: unreliable
- ✗ B. Guess the numbers: unreliable
- ✗ D. Ask an image generator to draw a chart that looks professional and has lots of colors: no real data

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-M-Q4 · question page 1559

AI.IMAGE.04 · Check generated diagrams, maps and charts against real data

What should generated images be used for?

Answer: D. Decoration, not evidence

Illustration only.

- ✗ A. Medical diagrams for patients: not reliable
- ✗ B. Official maps: not reliable
- ✗ C. Evidence in reports: not accurate

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-M-Q5 · question page 1560

AI.IMAGE.04 · Check generated diagrams, maps and charts against real data

A generated science diagram labels the heart as the lungs. What does this show?

Answer: B. Labels can be wrong

Check every label.

- ✗ A. That the tool is always right: not true
- ✗ C. That diagrams cannot have labels: they can
- ✗ D. That the heart and lungs are the same organ in some science textbooks used in schools: false

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-B1-Q1 · question page 1561

AI.IMAGE.04 · Check a chart against its numbers

A = 5, B = 10, C = 15. Which bar should be tallest?

Answer: C. C

Biggest number, tallest bar.

- ✗ A. B: middle value
- ✗ B. A, because the first bar in a chart is always drawn as the tallest one: no
- ✗ D. All equal: values differ

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-B1-Q2 · question page 1562

AI.IMAGE.04 · Check a chart against its numbers

Which should be shortest?

Answer: A. A

Smallest number.

- ✗ B. B: middle
- ✗ C. C, since the last bar in any chart is usually drawn the shortest of all: no
- ✗ D. None: one is smallest

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-B1-Q3 · question page 1563

AI.IMAGE.04 · Check a chart against its numbers

B = 10 and A = 5. How should B compare with A?

Answer: D. About twice as tall

10 is twice 5.

- ✗ A. Half as tall: reversed
- ✗ B. Ten times taller: too much
- ✗ C. About the same height: not proportional

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S75-B1-Q4 · question page 1564

AI.IMAGE.04 · Check a chart against its numbers

B is drawn shorter than A. What does that mean?

Answer: A. The chart is wrong

Heights must match numbers.

- ✗ B. A must be 20: label says 5
- ✗ C. B must be 3: label says 10
- ✗ D. The chart is correct: misleading

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-B1-Q5 · question page 1565

AI.IMAGE.04 · Check a chart against its numbers

What else should you read?

Answer: C. Whether the axis starts at 0

Axes can mislead.

- ✗ A. The chart's font: not the check
- ✗ B. The file name: not the check
- ✗ D. The color of each bar, because colors show the real values more clearly than heights: not the check

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-F1-Q1 · question page 1566

AI.IMAGE.04 · Use diagrams and charts responsibly in practice

Ana needs a map of bus routes for new students. What is the best source?

Answer: B. The transit agency's official map

Official, accurate source.

- ✗ A. An image generator asked to draw a detailed and accurate map of the city's bus routes: may invent routes
- ✗ C. A map of another city: wrong
- ✗ D. A sketch from memory: unreliable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-F1-Q2 · question page 1567

AI.IMAGE.04 · Use diagrams and charts responsibly in practice

A generated chart for a club report shows membership numbers. What should be checked?

Answer: D. Every number against club records

Compare with real data.

- ✗ A. Nothing: must check
- ✗ B. Only that the colors match the club logo, since the numbers come from the tool and are fine: numbers may be wrong
- ✗ C. Only the title: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-F1-Q3 · question page 1568

AI.IMAGE.04 · Use diagrams and charts responsibly in practice

A teacher wants a pretty background picture of planets for a slide title. Is a generated image reasonable?

Answer: B. Yes, as decoration labeled as an illustration

Decoration is fine.

- ✗ A. No, because generated images can never be used for anything at all in any classroom setting: too strong
- ✗ C. Only if it shows text: irrelevant
- ✗ D. Yes, and it can be used as a scientific diagram too: not accurate

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-F1-Q4 · question page 1569

AI.IMAGE.04 · Use diagrams and charts responsibly in practice

Which is the most accurate way to make a pie chart of a survey?

Answer: C. Enter the survey results into a spreadsheet chart

Real data in.

- ✗ A. Describe the survey to an image tool and ask for a pie chart that looks about right: not accurate
- ✗ B. Use a chart from another survey: wrong data
- ✗ D. Draw slices by guessing: not accurate

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-F1-Q5 · question page 1570

AI.IMAGE.04 · Use diagrams and charts responsibly in practice

A generated diagram of a plant has a part labeled "photosynthesis root." What should you do?

Answer: A. Check labels against a trusted source

Verify labels.

- ✗ B. Add more labels: more errors
- ✗ C. Make it bigger: not a fix
- ✗ D. Keep it, since diagrams always use correct scientific names even when they look unusual: not guaranteed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-F2-Q1 · question page 1571

AI.IMAGE.04 · Spot subtle chart and diagram errors

A chart's axis starts at 50, so 52 looks twice as tall as 51. What is the problem?

Answer: D. Small differences look huge

Truncated axes exaggerate.

- ✗ A. The numbers are wrong: the scale misleads
- ✗ B. The chart is too colorful: not the problem
- ✗ C. No problem: misleading

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-F2-Q2 · question page 1572

AI.IMAGE.04 · Spot subtle chart and diagram errors

A diagram is right except two labels are swapped. Why is it still a problem?

Answer: C. Readers learn the wrong names

Partly wrong is still wrong.

- ✗ A. Because diagrams need three labels: no
- ✗ B. Because swapped labels change the colors: no
- ✗ D. It is not a problem: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-F2-Q3 · question page 1573

AI.IMAGE.04 · Spot subtle chart and diagram errors

A map has the right shape but the wrong scale. What could go wrong?

Answer: A. Distances would be misjudged

Scale affects distance.

- ✗ B. Street names would change: not the issue
- ✗ C. The colors would fade: irrelevant
- ✗ D. Nothing: scale matters

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-F2-Q4 · question page 1574

AI.IMAGE.04 · Spot subtle chart and diagram errors

Why label a generated picture as "illustration" next to real data?

Answer: B. So readers do not treat it as accurate

Clear labeling.

- ✗ A. Because labels are required by every printer: no
- ✗ C. To make the file smaller: irrelevant
- ✗ D. So readers know the illustration is more accurate: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-F2-Q5 · question page 1575

AI.IMAGE.04 · Spot subtle chart and diagram errors

Which habit catches most chart errors?

Answer: D. Reading axes and legends and comparing with the source

Careful reading and comparison.

- ✗ A. Checking the font: not enough
- ✗ B. Counting the colors: not enough
- ✗ C. Looking at the overall shape of the chart for a second: misses errors

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-M-Q1 · question page 1577

AI.IMAGE.05 · Plan a presentation and tell a text draft from creating a file

What does a clear presentation structure include?

Answer: A. A purpose, main points with evidence, and a close

Purpose, points, evidence, close.

- ✗ B. Only pictures: missing structure
- ✗ C. Only a title slide: missing structure
- ✗ D. As many slides as possible, each packed with full paragraphs of text so nothing is left out: too much

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-M-Q2 · question page 1578

AI.IMAGE.05 · Plan a presentation and tell a text draft from creating a file

In this fictional example, what did the tool produce?

Fictional example (prepared in advance; not a real AI output)

Request:

Make me a 5-slide presentation on recycling.

Response:

Slide 1: Why Recycling Matters
Slide 2: What Can Be Recycled
Slide 3: How Our School Recycles
Slide 4: Common Mistakes
Slide 5: What You Can Do
(Each slide has 3 bullet points and speaker notes below.)

Answer: C. A text outline of slides

Text describing slides.

✗ A. A finished slide file saved on the computer that is ready to open in any presentation app: only text

✗ B. A video: no

✗ D. A printed handout: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-M-Q3 · question page 1579

AI.IMAGE.05 · Plan a presentation and tell a text draft from creating a file

How is an actual slide file usually made?

Answer: D. In a presentation app or with a real file-making feature

File creation is a separate step.

- ✗ A. By typing the word "slides" into a chat, which always creates a real file automatically: not always
- ✗ B. By printing the chat: not a slide file
- ✗ C. By renaming a text file: not a slide file

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-M-Q4 · question page 1580

AI.IMAGE.05 · Plan a presentation and tell a text draft from creating a file

What is a common rule for slides?

Answer: B. One idea per slide, few words

Simple, focused slides.

- ✗ A. Use full paragraphs: too wordy
- ✗ C. No titles: unclear
- ✗ D. Put every idea from the whole talk: crowded

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-M-Q5 · question page 1581

AI.IMAGE.05 · Plan a presentation and tell a text draft from creating a file

After a slide file is created, what should you do?

Answer: D. Open it and check every slide

Check the real file.

- ✗ A. Delete the outline: not needed
- ✗ B. Email it unopened: risky
- ✗ C. Present it straight away: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-B1-Q1 · question page 1582

AI.IMAGE.05 · Compare a text draft with a recipe versus a cooked meal

A text draft of slides is like what?

Answer: B. A recipe

Instructions, not the result.

- ✗ A. A grocery store: no
- ✗ C. A kitchen: no
- ✗ D. A cooked meal: not yet made

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-B1-Q2 · question page 1583

AI.IMAGE.05 · Compare a text draft with a recipe versus a cooked meal

The slide file is like what?

Answer: A. The cooked meal

The finished product.

- ✗ B. The oven: no
- ✗ C. The shopping list: no
- ✗ D. The recipe card: that is the draft

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-B1-Q3 · question page 1584

AI.IMAGE.05 · Compare a text draft with a recipe versus a cooked meal

Where is the meal made?

Answer: C. In a kitchen

Like a presentation app.

- ✗ A. At the table: no
- ✗ B. Inside the recipe book itself, as soon as someone reads the recipe out loud: no
- ✗ D. In the store: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-B1-Q4 · question page 1585

AI.IMAGE.05 · Compare a text draft with a recipe versus a cooked meal

Tasting the meal before serving is like what?

Answer: B. Opening and checking the file

Check the real result.

- ✗ A. Washing dishes: no
- ✗ C. Buying more food: no
- ✗ D. Reading the recipe one more time: not checking the result

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-B1-Q5 · question page 1586

AI.IMAGE.05 · Compare a text draft with a recipe versus a cooked meal

Can you present a recipe as if it were a meal?

Answer: C. No; it must be made first

Draft is not file.

- ✗ A. Only on holidays: no
- ✗ B. Yes, if it is long: no
- ✗ D. Yes, because a recipe and a meal are exactly the same thing when guests arrive: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-F1-Q1 · question page 1587

AI.IMAGE.05 · Plan and produce presentations in practice

Sam asks a chat for "slides" and gets bullet points in the chat window. What should he do next?

Answer: D. Build the slides in a presentation app from the text

Turn the draft into a file.

- ✗ A. Present the chat window: not slides
- ✗ B. Give up: unnecessary
- ✗ C. Look for the slide file in his downloads folder: may not exist

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-F1-Q2 · question page 1588

AI.IMAGE.05 · Plan and produce presentations in practice

A 10-minute talk has a 30-slide draft. What should change?

Answer: A. Cut it to about 5 to 10 slides

About one slide per minute or two.

- ✗ B. Remove the closing slide only: not enough
- ✗ C. Add 10 more slides: worse
- ✗ D. Keep all 30 slides: too many

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-F1-Q3 · question page 1589

AI.IMAGE.05 · Plan and produce presentations in practice

Which slide follows the "few words" rule?

Answer: D. Title "Recycling Saves Energy" with one chart

One idea, clear visual.

- ✗ A. A slide with 12 bullet points: too many
- ✗ B. A slide with only tiny text: hard to read
- ✗ C. A slide with four full paragraphs explaining every detail of recycling from start to finish: too wordy

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-F1-Q4 · question page 1590

AI.IMAGE.05 · Plan and produce presentations in practice

A tool with a real file feature makes a slide file. What should you check?

Answer: C. Every slide in the opened file

Check all slides.

- ✗ A. Only the first slide, because if the first one is fine then every other slide is fine too: not enough
- ✗ B. Nothing: must check
- ✗ D. Only the file name: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-F1-Q5 · question page 1591

AI.IMAGE.05 · Plan and produce presentations in practice

Which use of AI fits presentation planning best?

Answer: A. Drafting an outline you then revise

A starting point.

- ✗ B. Skipping preparation: not good
- ✗ C. Reading the AI text word for word: not yours
- ✗ D. Letting the tool decide the purpose: you decide

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-F2-Q1 · question page 1592

AI.IMAGE.05 · Check exported slides and make the talk your own

Slides look fine on a laptop but text overflows on the classroom screen. What should have happened?

Answer: B. Checking on the device that will show them

Test on the target device.

- ✗ A. Nothing, since every screen shows slides in exactly the same way regardless of the computer: they differ
- ✗ C. Using smaller images: not the issue
- ✗ D. Using more text: worse

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-F2-Q2 · question page 1593

AI.IMAGE.05 · Check exported slides and make the talk your own

Speaker notes say "studies show 90% of waste is recyclable." What should you do?

Answer: C. Check the claim before saying it

Notes are unverified answers.

- ✗ A. Delete all notes: unnecessary
- ✗ B. Make the number bigger: no
- ✗ D. Say it confidently, since speaker notes are written from verified sources by the tool: not guaranteed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-F2-Q3 · question page 1594

AI.IMAGE.05 · Check exported slides and make the talk your own

An exported file replaced your font and lost an image. What kind of problem is this?

Answer: D. An export problem

Exports can change files.

- ✗ A. A spelling problem: no
- ✗ B. A grounding problem: no
- ✗ C. A problem with the topic: not the topic

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-F2-Q4 · question page 1595

AI.IMAGE.05 · Check exported slides and make the talk your own

Why reorder and rewrite a generated outline?

Answer: A. So it fits your audience and time

Make it yours.

- ✗ B. So it becomes longer: not the goal
- ✗ C. Because teachers require rewriting: not the main reason
- ✗ D. Because generated outlines always contain nothing useful and must be thrown away: too strong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-F2-Q5 · question page 1596

AI.IMAGE.05 · Check exported slides and make the talk your own

About how many slides suit a 20-minute talk?

Answer: B. About 10 to 20

One slide per minute or two.

- ✗ A. Exactly 1: too few
- ✗ C. About 200: far too many
- ✗ D. About 80: too many

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S77-M-Q1 · question page 1598

AI.IMAGE.06 · Make visuals accessible with alt text, contrast and labels

What is alt text?

Answer: A. A short description so screen readers can read an image aloud

A description for accessibility.

- ✗ B. A caption in bold: not the same
- ✗ C. A hidden password: no
- ✗ D. The file name of the image, such as "photo123.jpg,": not useful

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-M-Q2 · question page 1599

AI.IMAGE.06 · Make visuals accessible with alt text, contrast and labels

Which alt text is best for a sales chart?

Answer: B. "Bar chart: sales doubled from March to June."

States what matters.

- ✗ A. "chart.png": useless
- ✗ C. "Image": useless
- ✗ D. "A chart image that someone made for this report.": misses the point

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-M-Q3 · question page 1600

AI.IMAGE.06 · Make visuals accessible with alt text, contrast and labels

What does good contrast mean?

Answer: D. Text stands out clearly from its background

Easy-to-see text.

- ✗ A. Text is very small: not contrast
- ✗ B. Text is in all capitals: not contrast
- ✗ C. Text uses many different bright colors: poor contrast

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-M-Q4 · question page 1601

AI.IMAGE.06 · Make visuals accessible with alt text, contrast and labels

Why put labels on chart lines instead of relying on color?

Answer: C. Some viewers cannot tell the colors apart

Color alone fails some people.

- ✗ A. To hide the data: no
- ✗ B. To make the chart smaller: not the reason
- ✗ D. Because colors are not allowed in any chart: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-M-Q5 · question page 1602

AI.IMAGE.06 · Make visuals accessible with alt text, contrast and labels

Can AI draft alt text?

Answer: A. Yes, but you check it

Draft, then verify.

- ✗ B. No, it cannot describe images: it can
- ✗ C. Yes, and its drafts never need checking, since tools describe images perfectly every time: they can be wrong
- ✗ D. Only for photos of food: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-B1-Q1 · question page 1603

AI.IMAGE.06 · Compare alt text with describing a picture over the phone

Saying "It's a file called IMG_204" is like what?

Answer: B. Useless alt text

Not a description.

- ✗ A. A chart label: no
- ✗ C. Good contrast: no
- ✗ D. Excellent alt text: not useful

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-B1-Q2 · question page 1604

AI.IMAGE.06 · Compare alt text with describing a picture over the phone

"Our team holding the trophy after the final" is like what?

Answer: C. Good alt text

What matters, briefly.

- ✗ A. Poor contrast: no
- ✗ B. A caption in bold: not the point
- ✗ D. A file name: it is a description

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-B1-Q3 · question page 1605

AI.IMAGE.06 · Compare alt text with describing a picture over the phone

Why skip the color of the wall?

Answer: D. It is not what matters

Focus on the purpose.

- ✗ A. Because walls are invisible: no
- ✗ B. To make it longer: no
- ✗ C. Because colors must never be mentioned in any description of any image at all: too strong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-B1-Q4 · question page 1606

AI.IMAGE.06 · Compare alt text with describing a picture over the phone

Who is alt text for?

Answer: B. People who cannot see the image

Screen reader users and others.

- ✗ A. Search engines only: not the main purpose
- ✗ C. Printers: no
- ✗ D. Only the person who created the image: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-B1-Q5 · question page 1607

AI.IMAGE.06 · Compare alt text with describing a picture over the phone

A good phone description is...

Answer: D. Short and focused on what matters

Brief and meaningful.

- ✗ A. A file name: useless
- ✗ B. Only one word: too little
- ✗ C. Very long, listing every single thing that appears anywhere in the picture: too much

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-F1-Q1 · question page 1608

AI.IMAGE.06 · Make visuals accessible in practice

Kim's slide uses yellow text on a white background. What should she change?

Answer: A. Use dark text on the light background

Improve contrast.

- ✗ B. Remove the text: loses content
- ✗ C. Add more yellow: worse
- ✗ D. Make the yellow text slightly larger: contrast still poor

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-F1-Q2 · question page 1609

AI.IMAGE.06 · Make visuals accessible in practice

A line chart uses red and green lines with no labels. What is the best fix?

Answer: C. Label each line with words

Labels beyond color.

- ✗ A. Use two slightly different shades of green for the lines so the chart looks more modern: worse
- ✗ B. Remove the legend: worse
- ✗ D. Make the chart smaller: worse

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-F1-Q3 · question page 1610

AI.IMAGE.06 · Make visuals accessible in practice

A tool drafts alt text "A group of people" for a photo of the science fair winners. What should you do?

Answer: A. Revise it to say who and why it matters

Check and improve drafts.

- ✗ B. Add the file name: not useful
- ✗ C. Keep it, since every description written by a tool is already as detailed as needed: misses the point
- ✗ D. Delete the alt text: worse

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-F1-Q4 · question page 1611

AI.IMAGE.06 · Make visuals accessible in practice

A newsletter image announces the event date only inside the picture. What helps screen reader users?

Answer: C. Also give the date as real text

Real text is readable.

- ✗ A. Move the picture lower: not a fix
- ✗ B. Add more pictures: not a fix
- ✗ D. Make the date in the picture bigger: they cannot read it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-F1-Q5 · question page 1612

AI.IMAGE.06 · Make visuals accessible in practice

Which alt text suits a decorative divider line?

Answer: B. Little or none, since it carries no information

Match alt text to purpose.

- ✗ A. "Important image": misleading
- ✗ C. The file name: not useful
- ✗ D. A detailed paragraph describing the line's exact color: unneeded

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-F2-Q1 · question page 1613

AI.IMAGE.06 · Fix weak alt text and color-only signals

Generated alt text for a chart says "colorful shapes." What is missing?

Answer: D. The data or key finding

Charts need their message.

- ✗ A. The file size: irrelevant
- ✗ B. The font name: irrelevant
- ✗ C. More colors, since describing the colors fully is the most important part of chart alt text: not the point

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-F2-Q2 · question page 1614

AI.IMAGE.06 · Fix weak alt text and color-only signals

A schedule marks late buses in red only. What improves it?

Answer: A. Add the word "late" or a symbol

Words or symbols, not only color.

- ✗ B. Remove the late buses: loses information
- ✗ C. Use pink instead: still color-only
- ✗ D. Make the red brighter, which ensures every reader can see that a bus is running late: still color-only

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-F2-Q3 · question page 1615

AI.IMAGE.06 · Fix weak alt text and color-only signals

Generated alt text mentions "a dog in the corner," but there is no dog. What happened?

Answer: C. The tool guessed something not there

Check drafts against the image.

- ✗ A. Alt text always includes animals: no
- ✗ B. The image changed: unlikely
- ✗ D. The dog must be hidden somewhere: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-F2-Q4 · question page 1616

AI.IMAGE.06 · Fix weak alt text and color-only signals

Should a publisher rely on screen-reader text recognition to make important text inside an image accessible?

Answer: A. No; provide real text or alt text

Some screen readers can recognize text in images, but it may be unavailable or wrong, so give important text as real text or alt text.

- ✗ B. Yes, if the image is large: size does not guarantee recognition
- ✗ C. Yes; screen readers read all text in every picture: some can recognize image text, but not reliably or everywhere
- ✗ D. Yes, if the words are in color: color does not make recognition reliable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-F2-Q5 · question page 1617

AI.IMAGE.06 · Fix weak alt text and color-only signals

What should alt text match?

Answer: B. The image's purpose

Purpose decides what to say.

- ✗ A. The image size: no
- ✗ C. The file name: no
- ✗ D. The length of the page: not the rule

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-M-Q1 · question page 1619

AI.MEDIA.01 · Tell transcription from speech generation and check transcripts

What does transcription do?

Answer: B. Turns spoken audio into written words

Speech to text.

- ✗ A. Deletes background noise only: not the definition
- ✗ C. Translates text into music: no
- ✗ D. Turns written words into a spoken voice: that is speech generation

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-M-Q2 · question page 1620

AI.MEDIA.01 · Tell transcription from speech generation and check transcripts

What does text-to-speech do?

Answer: D. Turns written words into spoken audio

Text to speech.

- ✗ A. Makes video from text: not this task
- ✗ B. Checks spelling: no
- ✗ C. Writes down what people say in a meeting: that is transcription

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-M-Q3 · question page 1621

AI.MEDIA.01 · Tell transcription from speech generation and check transcripts

What is a speaker label?

Answer: A. A note saying who said each part

Who said what.

- ✗ B. The file name: no
- ✗ C. The volume level: no
- ✗ D. A sticker on a loudspeaker: not the meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-M-Q4 · question page 1622

AI.MEDIA.01 · Tell transcription from speech generation and check transcripts

In this prepared example, which detail is most important to check against the recording?

Prepared example (a fictional meeting transcript)

Speaker 1: Our budget for the trip is fifteen hundred dollars.
Speaker 2: We can't book the bus until Friday.
Speaker 1: Then let's ask Mr. Patel to call the company.

Answer: C. "Fifteen hundred dollars" and "can't"

Numbers and negatives change meaning.

✗ A. The punctuation only: low stakes

✗ B. The word "Then," because short words at the start of sentences are the most often wrong: low stakes

✗ D. The word "the": low stakes

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-M-Q5 · question page 1623

AI.MEDIA.01 · Tell transcription from speech generation and check transcripts

"Wreck a nice beach" for "recognize speech" is what kind of error?

Answer: A. A misheard phrase

Similar sounds, wrong words.

- ✗ B. A missing word: not missing
- ✗ C. A text-to-speech error: it is transcription
- ✗ D. A speaker label error: not labeling

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-B1-Q1 · question page 1624

AI.MEDIA.01 · Compare transcription and speech generation with a court reporter and a narrator

A court reporter is like which task?

Answer: C. Transcription

Sound in, text out.

- ✗ A. Translation: no
- ✗ B. Image generation: no
- ✗ D. Speech generation: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-B1-Q2 · question page 1625

AI.MEDIA.01 · Compare transcription and speech generation with a court reporter and a narrator

An audiobook narrator is like which task?

Answer: D. Speech generation

Text in, sound out.

- ✗ A. Editing: no
- ✗ B. Summarizing: no
- ✗ C. Transcription: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-B1-Q3 · question page 1626

AI.MEDIA.01 · Compare transcription and speech generation with a court reporter and a narrator

What can a court reporter get wrong?

Answer: B. A name or who spoke

Like transcript errors.

- ✗ A. Nothing: they can err
- ✗ C. The judge's salary: irrelevant
- ✗ D. The color of the courtroom walls, which must be included in every court record: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-B1-Q4 · question page 1627

AI.MEDIA.01 · Compare transcription and speech generation with a court reporter and a narrator

For transcription, what goes in?

Answer: C. Sound

Audio input.

- ✗ A. Numbers only: no
- ✗ B. Pictures: no
- ✗ D. Written text: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-B1-Q5 · question page 1628

AI.MEDIA.01 · Compare transcription and speech generation with a court reporter and a narrator

For speech generation, what comes out?

Answer: D. Sound

Audio output.

- ✗ A. Written text that matches the words someone said in a recorded meeting earlier: reversed
- ✗ B. A spreadsheet: no
- ✗ C. A picture: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-F1-Q1 · question page 1629

AI.MEDIA.01 · Use and check transcripts in practice

Jo wants captions for a class video. Which task does she need?

Answer: B. Transcription

Speech becomes text.

- ✗ A. Image generation: no
- ✗ C. Music generation: no
- ✗ D. Text-to-speech: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-F1-Q2 · question page 1630

AI.MEDIA.01 · Use and check transcripts in practice

A blind student wants a news article read aloud. Which task fits?

Answer: A. Text-to-speech

Text becomes speech.

- ✗ B. Image tagging: no
- ✗ C. Video editing: no
- ✗ D. Transcription: it is already text

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-F1-Q3 · question page 1631

AI.MEDIA.01 · Use and check transcripts in practice

A transcript says the meeting agreed to "cancel" the event, but you recall "continue." What should you do?

Answer: C. Listen to that part of the recording

Check key statements.

- ✗ A. Ask attendees to vote again: check first
- ✗ B. Trust the transcript, since automatic transcripts never confuse words that sound alike: they can
- ✗ D. Delete the recording: loses evidence

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-F1-Q4 · question page 1632

AI.MEDIA.01 · Use and check transcripts in practice

A transcript credits the coach with a parent's question. What kind of error is this?

Answer: A. A wrong speaker label

Who said what.

- ✗ B. A missing word: no
- ✗ C. A speech generation error: no
- ✗ D. A misheard word: words fine

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-F1-Q5 · question page 1633

AI.MEDIA.01 · Use and check transcripts in practice

Which transcript items most need checking?

Answer: B. Names, numbers and decisions

High-stakes details.

- ✗ A. Greetings: low stakes
- ✗ C. Background music: not text
- ✗ D. Filler words like "um" and "uh,": low stakes

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-F2-Q1 · question page 1634

AI.MEDIA.01 · Anticipate when transcripts fail

Which recording is most likely to produce transcript errors?

Answer: D. A noisy room with people talking over each other

Noise and crosstalk.

- ✗ A. A short clear voicemail: easier
- ✗ B. A slow, scripted reading: easier
- ✗ C. A quiet room with one person speaking slowly: easiest case

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-F2-Q2 · question page 1635

AI.MEDIA.01 · Anticipate when transcripts fail

Why can "can" and "can't" be risky in transcripts?

Answer: B. They sound alike but mean the opposite

Small sound, big meaning.

- ✗ A. Because transcripts are not allowed to contain any short words with apostrophes: not the reason
- ✗ C. Because they are rare words: common
- ✗ D. Because they are spelled the same: they differ

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-F2-Q3 · question page 1636

AI.MEDIA.01 · Anticipate when transcripts fail

A transcript of a medical appointment lists a medicine dose. What should you do?

Answer: D. Confirm it with the provider or the recording

High stakes need confirmation.

- ✗ A. Double the dose to be safe: dangerous
- ✗ B. Follow the dose as written, since transcripts of appointments are checked by doctors: not guaranteed
- ✗ C. Ignore the dose: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-F2-Q4 · question page 1637

AI.MEDIA.01 · Anticipate when transcripts fail

Before recording a group conversation, what should you do?

Answer: A. Ask the people for permission

Consent rules vary by place.

- ✗ B. Record only the loud people: no
- ✗ C. Use a hidden phone: no
- ✗ D. Record secretly: rules differ

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-F2-Q5 · question page 1638

AI.MEDIA.01 · Anticipate when transcripts fail

A transcript misspells technical terms in a science lecture. Why?

Answer: C. Technical words are harder to recognize

Unfamiliar vocabulary.

- ✗ A. Because the file was too small: not the reason
- ✗ B. Because the lecturer must have spelled the words incorrectly while speaking them aloud: speech has no spelling
- ✗ D. Because science is not allowed: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-M-Q1 · question page 1640

AI.MEDIA.02 · Check subtitles and translated audio for timing, meaning and names

What are subtitles?

Answer: A. Spoken words shown as text on screen

On-screen text.

- ✗ B. Music under a video: no
- ✗ C. A voice in a second language that replaces the speaker's voice in the original video: that is dubbing
- ✗ D. The video's title only: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-M-Q2 · question page 1641

AI.MEDIA.02 · Check subtitles and translated audio for timing, meaning and names

Which is a timing problem?

Answer: C. A line appears after the person has finished speaking

Out of sync.

- ✗ A. A wrong place name: names
- ✗ B. A misspelled name: names
- ✗ D. A line that uses a word in the wrong language: meaning problem

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-M-Q3 · question page 1642

AI.MEDIA.02 · Check subtitles and translated audio for timing, meaning and names

In this prepared example, which subtitle has a meaning problem?

Prepared example (fictional generated subtitles)

Time	Original (English)	Generated subtitle (Spanish)
0:05	Break a leg tonight!	¡Rómpete una pierna esta noche!
0:09	The show starts at 7.	El espectáculo empieza a las 7.

Answer: B. The 0:05 line

The idiom was translated word for word.

- ✗ A. Neither line: one is literal
- ✗ C. Both lines: only one
- ✗ D. The 0:09 line: it is fine

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-M-Q4 · question page 1643

AI.MEDIA.02 · Check subtitles and translated audio for timing, meaning and names

Why can a fluent-sounding dub still be a problem?

Answer: D. It can change the meaning

Fluent is not faithful.

- ✗ A. Because dubs remove the video: no
- ✗ B. Because dubs are always too loud: not the issue
- ✗ C. Because fluent speech is always a sign: not always

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-M-Q5 · question page 1644

AI.MEDIA.02 · Check subtitles and translated audio for timing, meaning and names

Who should review important translated media?

Answer: C. A speaker of both languages

Bilingual review.

- ✗ A. Nobody, since translation tools are now as accurate as professional human translators in every case: not guaranteed
- ✗ B. The video's editor only: may not know both
- ✗ D. Someone who speaks only one language: cannot compare

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-B1-Q1 · question page 1645

AI.MEDIA.02 · Check a subtitle line with three questions

The subtitle appears two seconds late. Which check fails?

Answer: A. Timing

Out of sync.

- ✗ B. None: timing fails
- ✗ C. Names: not names
- ✗ D. Meaning: separate

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-B1-Q2 · question page 1646

AI.MEDIA.02 · Check a subtitle line with three questions

"Many animals are falling" fails which check?

Answer: D. Meaning

Idiom taken literally.

- ✗ A. None: meaning fails
- ✗ B. Names: not names
- ✗ C. Timing: not timing

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-B1-Q3 · question page 1647

AI.MEDIA.02 · Check a subtitle line with three questions

"Dr. Wen" for "Dr. Nguyen" fails which check?

Answer: B. Names

Wrong name.

- ✗ A. None: names fail
- ✗ C. Timing: not timing
- ✗ D. Meaning: not meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-B1-Q4 · question page 1648

AI.MEDIA.02 · Check a subtitle line with three questions

How many checks fail in the example?

Answer: D. Three

Timing, meaning and names.

- ✗ A. Two: all three
- ✗ B. None: all three
- ✗ C. One: all fail

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-B1-Q5 · question page 1649

AI.MEDIA.02 · Check a subtitle line with three questions

What should "raining cats and dogs" become?

Answer: A. A phrase meaning "raining very hard"

Translate the meaning.

- ✗ B. Nothing at all: loses meaning
- ✗ C. "It is sunny": wrong meaning
- ✗ D. A word-for-word version that keeps the cats and the dogs, so the subtitle is exact: loses meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-F1-Q1 · question page 1650

AI.MEDIA.02 · Review generated subtitles and dubs in practice

A school posts a video with generated Spanish subtitles for families. What is the best step before posting?

Answer: B. Have a Spanish speaker review them

Review important translations.

- ✗ A. Shorten the video: irrelevant
- ✗ C. Post right away, since every family will understand the subtitles even with small errors: may mislead
- ✗ D. Remove the English audio: unnecessary

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-F1-Q2 · question page 1651

AI.MEDIA.02 · Review generated subtitles and dubs in practice

Subtitles flash by too fast to read. Which check failed?

Answer: C. Timing

Lines must stay readable.

- ✗ A. None: timing failed
- ✗ B. Names: not names
- ✗ D. Meaning: separate

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-F1-Q3 · question page 1652

AI.MEDIA.02 · Review generated subtitles and dubs in practice

A dub translates "Mrs. Kowalski" as "Mrs. Cow." What failed?

Answer: D. Names

Names must be right.

- ✗ A. Timing, since the dubbed name was said slightly later than in the original video: not stated
- ✗ B. Only the volume: not the issue
- ✗ C. Nothing: it failed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-F1-Q4 · question page 1653

AI.MEDIA.02 · Review generated subtitles and dubs in practice

A subtitle says "5 miles" but the translated audio says "5 kilometers." What should you do?

Answer: B. Check which is right and fix the other

Units must match and be right.

- ✗ A. Delete the numbers: loses information
- ✗ C. Use the louder one: not a method
- ✗ D. Leave both, since miles and kilometers are about the same distance for most purposes: they differ

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-F1-Q5 · question page 1654

AI.MEDIA.02 · Review generated subtitles and dubs in practice

Which check is most important for a safety video?

Answer: C. Meaning of each instruction

Wrong meaning can cause harm.

- ✗ A. The video length: less important
- ✗ B. The music volume: less important
- ✗ D. The font of the subtitles: less important

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-F2-Q1 · question page 1655

AI.MEDIA.02 · Handle tone, consent and disclosure in translated media

A polite request in the original became a rude command in the dub. What changed?

Answer: A. The tone

Translation can shift politeness.

- ✗ B. The speaker's name: not names
- ✗ C. Nothing important: tone matters
- ✗ D. The timing: not timing

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-F2-Q2 · question page 1656

AI.MEDIA.02 · Handle tone, consent and disclosure in translated media

A dub uses a voice that imitates the original speaker. What question does this raise?

Answer: D. Whether the speaker consented

Imitated voices need consent.

- ✗ A. Whether the video has color: irrelevant
- ✗ B. Whether subtitles are needed: separate
- ✗ C. Whether the dub is long enough: not the issue

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-F2-Q3 · question page 1657

AI.MEDIA.02 · Handle tone, consent and disclosure in translated media

A team publishes machine subtitles that no one fully reviewed. What should they add?

Answer: C. A note that subtitles were machine-generated

Disclose unreviewed translation.

- ✗ A. Nothing, since viewers always assume that every subtitle they read has been reviewed: not true
- ✗ B. More subtitles: not disclosure
- ✗ D. A longer title: not disclosure

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-F2-Q4 · question page 1658

AI.MEDIA.02 · Handle tone, consent and disclosure in translated media

Why give viewers a way to report errors?

Answer: A. So mistakes can be found and fixed

Feedback improves accuracy.

- ✗ B. Because it is required for every video: not the reason
- ✗ C. To collect email addresses: not the goal
- ✗ D. So viewers can rewrite the entire video: not the goal

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-F2-Q5 · question page 1659

AI.MEDIA.02 · Handle tone, consent and disclosure in translated media

A joke in the original falls flat in translation. What is the best approach?

Answer: B. Have a reviewer adapt it

Adapt meaning, not just words.

- ✗ A. Remove the whole scene: too drastic
- ✗ C. Add laughter sounds: not a fix
- ✗ D. Translate it word for word anyway, since the exact words matter more than the joke: loses meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S80-M-Q1 · question page 1661

AI.MEDIA.03 · Explain voice cloning and why consent and disclosure matter

What is voice cloning?

Answer: C. Making a synthetic voice that sounds like a real person

Synthetic speech in someone's voice.

- ✗ A. Recording someone's voice once and playing back exactly the same recording many times: that is replay
- ✗ B. Making music from noise: no
- ✗ D. Turning speech into text: that is transcription

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

S80-M-Q2 · question page 1662

AI.MEDIA.03 · Explain voice cloning and why consent and disclosure matter

Which is a helpful use of voice cloning?

Answer: A. Helping someone who is losing their voice

Assistive use with consent.

- ✗ B. Faking a politician's statement: deceptive
- ✗ C. Making a fake celebrity endorsement: deceptive
- ✗ D. Calling a grandparent: a scam

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

S80-M-Q3 · question page 1663

AI.MEDIA.03 · Explain voice cloning and why consent and disclosure matter

What does disclosure mean here?

Answer: B. Telling listeners the voice is synthetic

Being open about it.

- ✗ A. Keeping secret that the voice was made by a computer so the audio sounds more natural: opposite
- ✗ C. Making the audio louder: no
- ✗ D. Publishing the person's address: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

S80-M-Q4 · question page 1664

AI.MEDIA.03 · Explain voice cloning and why consent and disclosure matter

What did Tennessee's ELVIS Act (effective July 1, 2024) do?

Answer: D. Added voice to that state's publicity rights

A state law on voice rights.

- ✗ A. Required all songs to be written by humans: not what it did
- ✗ B. Made voice cloning legal everywhere: not what it did
- ✗ C. Banned every use of synthetic voices anywhere: too broad

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

S80-M-Q5 · question page 1665

AI.MEDIA.03 · Explain voice cloning and why consent and disclosure matter

What did the FCC rule in February 2024?

Answer: C. AI voices in calls count as "artificial" voices

Existing artificial-voice rules, including their exceptions, apply to AI voices.

- ✗ A. AI voices may be used in any phone call to anyone: opposite
- ✗ B. Phone calls must be recorded: not the ruling
- ✗ D. Voice cloning is a crime everywhere: not the ruling

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

S80-B1-Q1 · question page 1666

AI.MEDIA.03 · Compare voice cloning with borrowing someone's signature

Why does a signature matter?

Answer: D. People believe you agreed when they see it

It stands for you.

- ✗ A. Because it is written in ink: not the point
- ✗ B. Because signatures are always the most beautiful part of any written document: not the reason
- ✗ C. Because it shows your height: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

S80-B1-Q2 · question page 1667

AI.MEDIA.03 · Compare voice cloning with borrowing someone's signature

Copying a signature without permission is like what?

Answer: C. Cloning a voice without consent

Unauthorized use.

- ✗ A. Writing your own name: no
- ✗ B. Typing a letter: no
- ✗ D. Getting written permission: that is fine

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

S80-B1-Q3 · question page 1668

AI.MEDIA.03 · Compare voice cloning with borrowing someone's signature

Using a cloned voice with consent and disclosure is like what?

Answer: A. Signing with permission and a note

Permitted and open.

- ✗ B. Erasing a signature: no
- ✗ C. Losing a pen: no
- ✗ D. Forging a signature perfectly: the opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

S80-B1-Q4 · question page 1669

AI.MEDIA.03 · Compare voice cloning with borrowing someone's signature

Why do listeners trust a familiar voice?

Answer: B. They believe the person is speaking

Voice signals identity.

- ✗ A. Because it is recorded: not the reason
- ✗ C. Because it is loud: no
- ✗ D. Because voices always tell the truth: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

S80-B1-Q5 · question page 1670

AI.MEDIA.03 · Compare voice cloning with borrowing someone's signature

Is a perfect copy acceptable without permission?

Answer: A. No; permission is what matters

Consent, not quality.

- ✗ B. Yes, if it is short: no
- ✗ C. Yes, because a copy that is perfect cannot cause any harm to the person whose voice it is: it can
- ✗ D. Only on weekends: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

S80-F1-Q1 · question page 1671

AI.MEDIA.03 · Apply consent and disclosure to voice cloning

A podcaster wants to use a cloned voice of a guest to fix a few words. What should come first?

Answer: C. The guest's clear permission

Consent before cloning.

- ✗ A. A louder microphone: irrelevant
- ✗ B. A longer episode: irrelevant
- ✗ D. Nothing, since small fixes to a few words never need anyone's permission at all: still their voice

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

S80-F1-Q2 · question page 1672

AI.MEDIA.03 · Apply consent and disclosure to voice cloning

An ad uses a synthetic voice that sounds like a famous singer without permission. What is the concern?

Answer: D. It uses the singer's identity without consent

Unauthorized use of identity.

- ✗ A. The ad is too short: irrelevant
- ✗ B. The ad has music: irrelevant
- ✗ C. None, since a synthetic voice is a new voice: it imitates someone

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

S80-F1-Q3 · question page 1673

AI.MEDIA.03 · Apply consent and disclosure to voice cloning

A school announcement uses a synthetic voice reading the principal's message, with permission. What should it add?

Answer: C. A note that the voice is synthetic

Disclosure.

- ✗ A. Nothing, since permission from the principal means listeners never need to be told anything: disclosure still helps
- ✗ B. A different message: not needed
- ✗ D. Louder music: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

S80-F1-Q4 · question page 1674

AI.MEDIA.03 · Apply consent and disclosure to voice cloning

Grandpa gets a call in his grandson's voice asking for money urgently. What should he do?

Answer: B. Hang up and call his grandson on a known number

Verify through a known channel.

- ✗ A. Give his bank details: unsafe
- ✗ C. Send the money right away, since the voice sounded exactly like his grandson: voices can be cloned
- ✗ D. Call back the number that called: may be the scammer

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

S80-F1-Q5 · question page 1675

AI.MEDIA.03 · Apply consent and disclosure to voice cloning

A company makes non-emergency telemarketing robocalls to cellphones with an AI voice and no prior consent. Under the FCC's 2024 ruling, what applies?

Answer: D. The same consent rules as other artificial-voice calls

Covered AI-voice calls follow the same consent rules as other artificial-voice calls.

- ✗ A. A ban on all phone calls: too broad
- ✗ B. Only a fee: not stated
- ✗ C. No rules, because AI voices were left out of phone-call rules entirely by the ruling: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

S80-F2-Q1 · question page 1676

AI.MEDIA.03 · Navigate differing laws and protect against voice scams

A voice use is allowed in one state. Is it allowed everywhere?

Answer: A. Not necessarily; laws differ by place

Check local law.

- ✗ B. Yes, if it is online: no
- ✗ C. No, it is banned everywhere: too strong
- ✗ D. Yes, because every state and every country follows exactly the same rules about voices: they differ

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

S80-F2-Q2 · question page 1677

AI.MEDIA.03 · Navigate differing laws and protect against voice scams

What is a practical rule everywhere?

Answer: B. Get clear permission before cloning a voice

Permission first.

- ✗ A. Only clone famous people: worse
- ✗ C. Never tell anyone a voice is synthetic: opposite
- ✗ D. Clone any voice you like: still needs consent

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

S80-F2-Q3 · question page 1678

AI.MEDIA.03 · Navigate differing laws and protect against voice scams

How can a family protect against fake voice calls?

Answer: C. Agree on a code word to verify each other

A shared secret.

- ✗ A. Post their voices online: risky
- ✗ B. Answer every call quickly and do whatever the familiar voice asks without any delay: risky
- ✗ D. Stop using phones: impractical

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

S80-F2-Q4 · question page 1679

AI.MEDIA.03 · Navigate differing laws and protect against voice scams

Are all legal questions about voice cloning settled?

Answer: A. No; courts are still deciding many

An evolving area.

- ✗ B. There are no laws about voices: some exist
- ✗ C. Yes, by the ELVIS Act for the whole world: one state
- ✗ D. Yes; every possible question about synthetic voices was answered by courts years ago: still developing

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

S80-F2-Q5 · question page 1680

AI.MEDIA.03 · Navigate differing laws and protect against voice scams

Which use most clearly crosses the line?

Answer: D. Imitating a real person to deceive listeners

Deception is the problem.

- ✗ A. A clearly labeled synthetic narrator: acceptable
- ✗ B. A screen reader voice: acceptable
- ✗ C. Using a synthetic voice, with consent: acceptable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

S81-M-Q1 · question page 1682

AI.MEDIA.04 · Check the rights and terms that apply before using generated music

What should you check first in a music tool's terms?

Answer: C. Whether commercial use is allowed

Use rights.

- ✗ A. The tool's logo: irrelevant
- ✗ B. The number of users: irrelevant
- ✗ D. The color scheme of the app: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-M-Q2 · question page 1683

AI.MEDIA.04 · Check the rights and terms that apply before using generated music

Per the US Copyright Office's January 2025 report, do prompts alone make you the author?

Answer: B. No

Prompts alone are not enough.

- ✗ A. Yes, because writing a detailed prompt always makes you the full author of every output: not per the report
- ✗ C. Only for songs under 3 minutes: no
- ✗ D. Only for long prompts: not per the report

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-M-Q3 · question page 1684

AI.MEDIA.04 · Check the rights and terms that apply before using generated music

Which can be protected by copyright, per that report?

Answer: A. Your own lyrics or meaningful edits

Human-authored contributions.

- ✗ B. Any random sound: no
- ✗ C. Someone else's melody: not yours
- ✗ D. The tool's output, in full, simply: not per the report

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-M-Q4 · question page 1685

AI.MEDIA.04 · Check the rights and terms that apply before using generated music

A generated track sounds almost exactly like a famous song. What should you do?

Answer: D. Avoid using it

Similarity risk.

- ✗ A. Speed it up slightly: not enough
- ✗ B. Change the title: not enough
- ✗ C. Use it: they can be similar

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-M-Q5 · question page 1686

AI.MEDIA.04 · Check the rights and terms that apply before using generated music

What record should you keep?

Answer: B. The tool, plan and date

Proof of terms at the time.

- ✗ A. Nothing: helps to keep records
- ✗ C. The tool's home page design: irrelevant
- ✗ D. Only the song's title: may change

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-B1-Q1 · question page 1687

AI.MEDIA.04 · Compare music rights with borrowing a library book

What rules come with a library book?

Answer: A. You can read it but not sell copies

Use limits.

- ✗ B. You must memorize it: no
- ✗ C. You must read it in one day: not the rule
- ✗ D. You can do anything at all with it: not allowed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-B1-Q2 · question page 1688

AI.MEDIA.04 · Compare music rights with borrowing a library book

In music tools, what sets the rules?

Answer: D. The tool's terms

Terms decide use.

- ✗ A. The song's length: not the source of rules
- ✗ B. The weather: no
- ✗ C. The listener's mood: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-B1-Q3 · question page 1689

AI.MEDIA.04 · Compare music rights with borrowing a library book

Writing your name in a library book is like what?

Answer: C. Typing a prompt

Not authorship by itself.

- ✗ A. Composing the music yourself from scratch, note by note, without any tool at all: that is authorship
- ✗ B. Buying the book: no
- ✗ D. Returning the book: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-B1-Q4 · question page 1690

AI.MEDIA.04 · Compare music rights with borrowing a library book

A free plan allowing only personal use is like what?

Answer: D. Reading a library book at home

Limited use.

- ✗ A. Owning the library: no
- ✗ B. Writing a new book: no
- ✗ C. Photocopying the library book: not allowed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-B1-Q5 · question page 1691

AI.MEDIA.04 · Compare music rights with borrowing a library book

What is the lesson?

Answer: A. Check the rules before using generated music

Read the terms.

- ✗ B. Generated music has no rules at all, so it can be used anywhere for anything: it has terms
- ✗ C. Never listen to music: no
- ✗ D. Only use library music: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-F1-Q1 · question page 1692

AI.MEDIA.04 · Use generated music responsibly in practice

Eli wants generated background music for a monetized video. What should he check?

Answer: C. That his plan allows commercial use

Commercial use rights.

- ✗ A. Nothing, because all generated music may be used in any video, including monetized ones: depends on terms
- ✗ B. Only the song length: not enough
- ✗ D. Only the tempo: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-F1-Q2 · question page 1693

AI.MEDIA.04 · Use generated music responsibly in practice

A student writes her own lyrics and uses a tool for the beat. Which part is most clearly hers?

Answer: B. The lyrics

Human-authored contribution.

- ✗ A. The tool's code: no
- ✗ C. Neither: the lyrics are hers
- ✗ D. The generated beat: prompts alone are not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-F1-Q3 · question page 1694

AI.MEDIA.04 · Use generated music responsibly in practice

A prompt asks for a song "in the voice of" a named living singer. What is the concern?

Answer: D. Imitating a real artist

Avoid imitating real artists.

- ✗ A. The song may be too slow: irrelevant
- ✗ B. The song may be too short: irrelevant
- ✗ C. None: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-F1-Q4 · question page 1695

AI.MEDIA.04 · Use generated music responsibly in practice

A tool's terms say "credit required." What should you do?

Answer: C. Include the credit as described

Follow the terms.

- ✗ A. Credit yourself only: not what is required
- ✗ B. Change tools after publishing: too late
- ✗ D. Leave out the credit: required

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-F1-Q5 · question page 1696

AI.MEDIA.04 · Use generated music responsibly in practice

Why save proof of your plan when you generated a track?

Answer: B. Terms can change later

Records protect you.

- ✗ A. Because it is required by every country: not the reason
- ✗ C. Because tools delete every song that was made on a free plan after one week: not the reason
- ✗ D. To make the song louder: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-F2-Q1 · question page 1697

AI.MEDIA.04 · Reduce rights risks while the law develops

Are rights questions about generated music fully settled?

Answer: A. No; they are still being argued

An open area.

- ✗ B. There are no rules: there are
- ✗ C. Yes, the same in every country: they differ
- ✗ D. Yes; one court case: not settled

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-F2-Q2 · question page 1698

AI.MEDIA.04 · Reduce rights risks while the law develops

Which terms should you rely on?

Answer: C. The current terms

Read the current version.

- ✗ A. Any terms you like: no
- ✗ B. A friend's memory of the terms: unreliable
- ✗ D. Terms from a blog post written several years ago about a different version of the tool: may be outdated

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-F2-Q3 · question page 1699

AI.MEDIA.04 · Reduce rights risks while the law develops

A platform flags your video because its generated music matches an existing recording. What does this show?

Answer: B. Generated music can resemble existing works

Similarity can be detected.

- ✗ A. That videos need no music: no
- ✗ C. That the platform is always wrong, since generated songs cannot match existing recordings: they can
- ✗ D. That music is banned: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-F2-Q4 · question page 1700

AI.MEDIA.04 · Reduce rights risks while the law develops

For a school concert, what should guide the music choice?

Answer: D. The venue's rules on music

Follow the venue's rules.

- ✗ A. The loudest song: not a rule
- ✗ B. Whatever is trending: not a rule
- ✗ C. Only what the tool allows: they can

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-F2-Q5 · question page 1701

AI.MEDIA.04 · Reduce rights risks while the law develops

What does this course do on disputed music rights?

Answer: A. It does not settle them

Disputed questions stay open.

- ✗ B. It says all generated music is illegal: not the position
- ✗ C. It gives one final answer that applies in every country for every kind of song: it does not
- ✗ D. It says anything goes: not the position

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S82-M-Q1 · question page 1703

AI.MEDIA.05 · Check video for consistency over time and disclose synthetic presenters

What challenge does video add compared with single images?

Answer: A. Consistency over time

Frames must stay consistent.

- ✗ B. Videos have no people: no
- ✗ C. Videos have no colors: no
- ✗ D. Video can never be generated by AI tools, so it is always real footage from a camera: it can be

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-M-Q2 · question page 1704

AI.MEDIA.05 · Check video for consistency over time and disclose synthetic presenters

Which is a sign of a generated video?

Answer: C. A shirt that changes pattern between frames

Inconsistency over time.

- ✗ A. A video with music: not a sign
- ✗ B. A short video: not a sign
- ✗ D. A video that was recorded in a bright room: not a sign

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-M-Q3 · question page 1705

AI.MEDIA.05 · Check video for consistency over time and disclose synthetic presenters

What should you check about lip movement?

Answer: B. Whether it matches the words

Lip sync.

- ✗ A. Whether the person has a microphone: not the check
- ✗ C. Whether the person is smiling: not the check
- ✗ D. Whether the lips are a natural color: not the main check

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-M-Q4 · question page 1706

AI.MEDIA.05 · Check video for consistency over time and disclose synthetic presenters

What is a synthetic presenter?

Answer: D. An AI-generated person or avatar presenting a video

A generated presenter.

- ✗ A. A microphone: no
- ✗ B. A video file format: no
- ✗ C. A real news anchor who reads the news from a teleprompter: real person

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-M-Q5 · question page 1707

AI.MEDIA.05 · Check video for consistency over time and disclose synthetic presenters

How should a synthetic presenter be disclosed?

Answer: A. With an on-screen label or description note

Tell viewers clearly.

- ✗ B. It does not need disclosing if it looks realistic enough that viewers will not notice: opposite
- ✗ C. By adding music: not disclosure
- ✗ D. By making the video shorter: not disclosure

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-B1-Q1 · question page 1708

AI.MEDIA.05 · Compare video consistency with continuity in movies

What is the continuity job in movies?

Answer: C. Keeping details the same between shots

Consistent details.

- ✗ A. Selling tickets: no
- ✗ B. Choosing actors: no
- ✗ D. Writing the music that plays during the final scenes of the movie for the audience: not continuity

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-B1-Q2 · question page 1709

AI.MEDIA.05 · Compare video consistency with continuity in movies

A coffee cup half full, then full in the next shot, is what?

Answer: D. A continuity slip

Details changed.

- ✗ A. Good editing: it is a slip
- ✗ B. A lighting choice: no
- ✗ C. A clever special effect: not intended

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-B1-Q3 · question page 1710

AI.MEDIA.05 · Compare video consistency with continuity in movies

A necklace disappearing in a generated video is like what?

Answer: B. A continuity slip

Inconsistency.

- ✗ A. A sound error: no
- ✗ C. A subtitle error: no
- ✗ D. Perfect continuity: not realistic

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-B1-Q4 · question page 1711

AI.MEDIA.05 · Compare video consistency with continuity in movies

Who notices continuity slips?

Answer: A. Viewers

They are visible.

- ✗ B. Only the director: viewers too
- ✗ C. Only the camera: no
- ✗ D. Nobody: they notice

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-B1-Q5 · question page 1712

AI.MEDIA.05 · Compare video consistency with continuity in movies

What should you look for in generated video?

Answer: B. Details changing between moments

Continuity checks.

- ✗ A. Only the length: not enough
- ✗ C. Only the music: not enough
- ✗ D. Only the title at the start: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-F1-Q1 · question page 1713

AI.MEDIA.05 · Check and disclose video in practice

A training video uses an AI avatar to present safety rules. What should it include?

Answer: C. A clear note that the presenter is AI-generated

Disclose synthetic presenters.

- ✗ A. Nothing extra, since viewers only care about the safety rules and not about who presents them: disclosure matters
- ✗ B. A fake name and job title: deceptive
- ✗ D. A claim the presenter is real: deceptive

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-F1-Q2 · question page 1714

AI.MEDIA.05 · Check and disclose video in practice

In a clip, a person's earrings vanish and return. What does this suggest?

Answer: D. Possible generation or editing

Inconsistency is a clue.

- ✗ A. A bad microphone: irrelevant
- ✗ B. A slow internet connection: not likely
- ✗ C. Nothing: not normal

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-F1-Q3 · question page 1715

AI.MEDIA.05 · Check and disclose video in practice

A video's words do not match the speaker's lip movements. What should you consider?

Answer: C. The audio may have been changed or generated

Lip sync mismatch.

- ✗ A. Nothing: worth checking
- ✗ B. That the video is very old: not necessarily
- ✗ D. That the speaker has a cold, which always makes lips move differently from the words: not likely

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-F1-Q4 · question page 1716

AI.MEDIA.05 · Check and disclose video in practice

A ball passes through a wall in a video meant to look real. What does this break?

Answer: A. Ordinary physics

Physics errors.

- ✗ B. Lip sync, since the ball and the wall are both moving their lips in the scene: no
- ✗ C. Subtitles: no
- ✗ D. Nothing: it is impossible

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-F1-Q5 · question page 1717

AI.MEDIA.05 · Check and disclose video in practice

A small business makes an ad with a synthetic spokesperson. What is required by honesty?

Answer: D. Telling viewers the spokesperson is synthetic

Disclosure.

- ✗ A. Using more music: irrelevant
- ✗ B. Hiding the business name: irrelevant
- ✗ C. Saying the spokesperson is a real customer: deceptive

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-F2-Q1 · question page 1718

AI.MEDIA.05 · Handle subtle edits and context in video

A real video has one word changed by AI. Why can this matter?

Answer: B. It can change the meaning

Small edits, big meaning.

- ✗ A. It changes the colors: irrelevant
- ✗ C. It makes the file bigger: irrelevant
- ✗ D. It cannot matter: it can

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-F2-Q2 · question page 1719

AI.MEDIA.05 · Handle subtle edits and context in video

A clip is cut so a person seems to say the opposite of what they meant. What is the problem?

Answer: A. Removed context misleads

Context matters.

- ✗ B. The clip is too long: not the issue
- ✗ C. The clip has no music: not the issue
- ✗ D. No problem: context matters

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-F2-Q3 · question page 1720

AI.MEDIA.05 · Handle subtle edits and context in video

When publishing an edited video, what should you keep?

Answer: B. The original footage

Originals show what changed.

- ✗ A. Nothing: keep originals
- ✗ C. Only the audio: not enough
- ✗ D. Only the edited version: keep it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-F2-Q4 · question page 1721

AI.MEDIA.05 · Handle subtle edits and context in video

A synthetic presenter looks like a real local teacher. What is needed?

Answer: D. The teacher's consent

Likeness needs consent.

- ✗ A. A different background: irrelevant
- ✗ B. A longer video: irrelevant
- ✗ C. Nothing: consent needed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-F2-Q5 · question page 1722

AI.MEDIA.05 · Handle subtle edits and context in video

Which use should never happen?

Answer: D. Making a real person seem to endorse a product without approval

An unapproved endorsement presented as genuine deceives viewers and misuses a real identity.

- ✗ A. Replacing a plain background with a nicer one and noting the change in the video description: acceptable
- ✗ B. Adding captions: acceptable
- ✗ C. Trimming silence: acceptable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S83-M-Q1 · question page 1724

AI.MEDIA.06 · Verify surprising media by checking source and context

What is a deepfake?

Answer: D. Realistic fake media showing someone doing what they did not

Realistic and false.

- ✗ A. A cartoon: no
- ✗ B. A very old photo: no
- ✗ C. Any video that has been edited at all, including trimming silence: too broad

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

S83-M-Q2 · question page 1725

AI.MEDIA.06 · Verify surprising media by checking source and context

Why is looking for one visual oddity unreliable?

Answer: B. Real videos have glitches and new fakes may have none

Looks can mislead both ways.

- ✗ A. Because oddities are always invisible: no
- ✗ C. Because all videos are fake: no
- ✗ D. Because fakes always show strange blinking, so if blinking looks normal the video is real: not reliable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

S83-M-Q3 · question page 1726

AI.MEDIA.06 · Verify surprising media by checking source and context

Which is a source check?

Answer: C. Finding who first posted it

Origin matters.

- ✗ A. Looking at the thumbnail: not source
- ✗ B. Counting how many times the person in the video blinks during the first ten seconds: visual, not source
- ✗ D. Checking the video length: not source

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

S83-M-Q4 · question page 1727

AI.MEDIA.06 · Verify surprising media by checking source and context

What can a reverse image search find?

Answer: A. Older, original versions of an image

Earlier versions.

- ✗ B. The video's future views: no
- ✗ C. The camera's battery level: no
- ✗ D. The exact person who made a fake, along with their home address and phone number: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

S83-M-Q5 · question page 1728

AI.MEDIA.06 · Verify surprising media by checking source and context

Which clip deserves the most checking?

Answer: B. A surprising one that makes you angry or scared

Strong emotions, more checks.

- ✗ A. A weather report: low risk
- ✗ C. A calm, ordinary clip of a local park on a sunny afternoon with nothing much happening: low risk
- ✗ D. A cooking video: low risk

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

S83-B1-Q1 · question page 1729

AI.MEDIA.06 · Verify a clip with four questions

A clip posted by a brand-new anonymous account fails which question?

Answer: C. Source

Untrustworthy origin.

- ✗ A. Earlier versions: not shown
- ✗ B. Confirmation: not shown
- ✗ D. Context: it is source

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

S83-B1-Q2 · question page 1730

AI.MEDIA.06 · Verify a clip with four questions

Snow on a 35 °C day fails which question?

Answer: A. Context

Facts do not fit.

- ✗ B. Confirmation: not shown
- ✗ C. Earlier versions: not shown
- ✗ D. Source: it is context

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

S83-B1-Q3 · question page 1731

AI.MEDIA.06 · Verify a clip with four questions

No trusted outlet reports a "major" event. Which question fails?

Answer: D. Confirmation

No confirmation.

- ✗ A. Context: separate
- ✗ B. Source: separate
- ✗ C. Earlier versions: not shown

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

S83-B1-Q4 · question page 1732

AI.MEDIA.06 · Verify a clip with four questions

A reverse search finds the same photo from 2018 with a different caption. Which question?

Answer: B. Earlier versions

Recycled media.

- ✗ A. Context: separate
- ✗ C. Source: separate
- ✗ D. Confirmation: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

S83-B1-Q5 · question page 1733

AI.MEDIA.06 · Verify a clip with four questions

How many of the four questions does the snow example fail?

Answer: D. Two

Source and context.

- ✗ A. One: two
- ✗ B. None: two
- ✗ C. Four: only two shown

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

S83-F1-Q1 · question page 1734

AI.MEDIA.06 · Verify surprising media in practice

A video shows a mayor announcing schools will close tomorrow. What should you check first?

Answer: C. The city's official accounts or trusted news

Official confirmation.

- ✗ A. The video's music: irrelevant
- ✗ B. How many likes it has: popularity is not proof
- ✗ D. Whether the mayor blinks normally in the video: unreliable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

S83-F1-Q2 · question page 1735

AI.MEDIA.06 · Verify surprising media in practice

A dramatic photo of a flood is going viral. A reverse search finds it from another country years ago. What does this show?

Answer: A. It is being shared with false context

Real image, false context.

- ✗ B. Nothing: it shows misuse
- ✗ C. That the photo is AI-generated: it is real but recycled
- ✗ D. That the flood is happening again: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

S83-F1-Q3 · question page 1736

AI.MEDIA.06 · Verify surprising media in practice

A friend shares a shocking audio clip of a celebrity. The clip has no source. What is the best response?

Answer: B. Do not share it until it is confirmed

Verify before sharing.

- ✗ A. Report the friend to police: excessive
- ✗ C. Add your own comments and repost: spreads it
- ✗ D. Share it with a note saying "might be fake,": still spreads

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

S83-F1-Q4 · question page 1737

AI.MEDIA.06 · Verify surprising media in practice

Which detail is the strongest sign a clip is genuine?

Answer: D. Several trusted outlets confirm it independently

Independent confirmation.

- ✗ A. It was posted recently: not proof
- ✗ B. It has many views: not proof
- ✗ C. The clip looks very high quality and sharp: quality is not proof

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

S83-F1-Q5 · question page 1738

AI.MEDIA.06 · Verify surprising media in practice

A clip shows a protest "today," but trees are bare and it is summer. Which check does this use?

Answer: C. Context

Season does not fit.

- ✗ A. Earlier versions: not this
- ✗ B. Confirmation: not this
- ✗ D. Source: not source

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

S83-F2-Q1 · question page 1739

AI.MEDIA.06 · Avoid both mistakes and respond to harmful fakes

What are the two common mistakes?

Answer: A. Believing fakes and dismissing real evidence as AI

Both errors.

- ✗ B. Using reverse search and reading captions: good habits
- ✗ C. Watching videos and reading articles: not mistakes
- ✗ D. Checking sources too carefully and asking trusted news outlets whether events really happened: good habits

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

S83-F2-Q2 · question page 1740

AI.MEDIA.06 · Avoid both mistakes and respond to harmful fakes

Someone says a real recording of them is "probably AI" to avoid blame. What is this an example of?

Answer: B. Dismissing real evidence

Denying real media.

- ✗ A. Disclosure: no
- ✗ C. Good fact-checking: not checking
- ✗ D. Believing a fake: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

S83-F2-Q3 · question page 1741

AI.MEDIA.06 · Avoid both mistakes and respond to harmful fakes

What avoids both mistakes?

Answer: A. Checking sources rather than trusting looks

Source-based checking.

- ✗ B. Trusting your first impression of how real a video looks, without any further checks: causes both
- ✗ C. Never watching videos: impractical
- ✗ D. Believing whatever is most popular: unreliable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

S83-F2-Q4 · question page 1742

AI.MEDIA.06 · Avoid both mistakes and respond to harmful fakes

From May 19, 2026, what must covered US platforms do under the TAKE IT DOWN Act?

Answer: C. Offer a removal process and act on valid requests within 48 hours

A dated federal requirement.

- ✗ A. Nothing new: it adds duties
- ✗ B. Ban all video uploads: no
- ✗ D. Remove every AI-generated image from their sites, including art: too broad

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

S83-F2-Q5 · question page 1743

AI.MEDIA.06 · Avoid both mistakes and respond to harmful fakes

You receive a nonconsensual intimate deepfake of a classmate. What should you do?

Answer: D. Report it and do not share it

Report, do not spread.

- ✗ A. Keep it to show later: harmful
- ✗ B. Post it with a warning: spreads harm
- ✗ C. Forward it to a few friends: spreads harm

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

S84-M-Q1 · question page 1745

AI.MEDIA.07 · Explain provenance labels and the limits of detection

What does provenance mean?

Answer: A. A record of where content came from and how it changed

Origin and history.

- ✗ B. The number of views: no
- ✗ C. A rating of how beautiful or well made a picture is compared with other pictures online: not the meaning
- ✗ D. The file size: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S84-M-Q2 · question page 1746

AI.MEDIA.07 · Explain provenance labels and the limits of detection

What are Content Credentials based on?

Answer: C. The C2PA standard

Coalition for Content Provenance and Authenticity.

- ✗ A. A social media app: no
- ✗ B. A camera brand: no
- ✗ D. A single government law: it is a standard

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S84-M-Q3 · question page 1747

AI.MEDIA.07 · Explain provenance labels and the limits of detection

How can provenance records be lost?

Answer: B. By screenshots or re-uploading

Records can be stripped.

- ✗ A. By looking at it: no
- ✗ C. By opening the file: not usually
- ✗ D. They can never be lost once they are attached, no matter what happens to the file: they can

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S84-M-Q4 · question page 1748

AI.MEDIA.07 · Explain provenance labels and the limits of detection

An image has no provenance label. What does that prove?

Answer: D. Nothing about whether it is real or fake

Missing labels are not proof.

- ✗ A. That it is real: not proven
- ✗ B. That it is old: not proven
- ✗ C. That the image must be fake, because every real photo always carries a provenance label: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S84-M-Q5 · question page 1749

AI.MEDIA.07 · Explain provenance labels and the limits of detection

What do detector tools give?

Answer: B. Scores that can be wrong both ways

Detector scores are estimates, often shown as percentages that are not true probabilities, and they make errors both ways.

- ✗ A. Certain answers that prove beyond doubt: not certain
- ✗ C. A copyright license: no
- ✗ D. The creator's name: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S84-B1-Q1 · question page 1750

AI.MEDIA.07 · Compare provenance with a package tracking history

What does a tracking history show?

Answer: D. Where the parcel has been

Its history.

- ✗ A. The driver's name only: not the main point
- ✗ B. The weather: no
- ✗ C. What is inside the parcel: not the purpose

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S84-B1-Q2 · question page 1751

AI.MEDIA.07 · Compare provenance with a package tracking history

Provenance for a file is like what?

Answer: A. A tracking history

Origin and changes.

- ✗ B. A gift card: no
- ✗ C. A price tag: no
- ✗ D. A brand-new plain box: that is missing provenance

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S84-B1-Q3 · question page 1752

AI.MEDIA.07 · Compare provenance with a package tracking history

Repacking the item in a plain box is like what?

Answer: C. Stripping provenance by re-uploading

Records lost.

- ✗ A. Delivering faster: no
- ✗ B. Opening the parcel: no
- ✗ D. Adding a new and more detailed tracking label that shows the parcel's full history: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S84-B1-Q4 · question page 1753

AI.MEDIA.07 · Compare provenance with a package tracking history

A box with no tracking proves what?

Answer: B. Not necessarily anything

Absence is not proof.

- ✗ A. That it is valuable: no
- ✗ C. That the box was definitely stolen and should be reported to the police at once: not proof
- ✗ D. That it is empty: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S84-B1-Q5 · question page 1754

AI.MEDIA.07 · Compare provenance with a package tracking history

What is the lesson?

Answer: C. Provenance helps but its absence proves little

Useful but limited.

- ✗ A. Only buy online: no
- ✗ B. Tracking is useless: too strong
- ✗ D. Never trust anything that has a tracking history: not the lesson

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S84-F1-Q1 · question page 1755

AI.MEDIA.07 · Use provenance and detection carefully in practice

A news photo shows Content Credentials listing the camera and edits. What does this give you?

Answer: A. Useful information about its history

A helpful clue.

- ✗ B. A copyright license: no
- ✗ C. Nothing useful: it helps
- ✗ D. Absolute proof that everything shown in the photo is true, so no other checks are needed: still check context

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S84-F1-Q2 · question page 1756

AI.MEDIA.07 · Use provenance and detection carefully in practice

A viral image has no credentials because it was screenshotted. What should you conclude?

Answer: D. Nothing yet; check other clues

Absence proves little.

- ✗ A. That it is fake, because only fake images lose their credentials when they are shared: not proof
- ✗ B. That it is old: not proven
- ✗ C. That it is real: not proven

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S84-F1-Q3 · question page 1757

AI.MEDIA.07 · Use provenance and detection carefully in practice

A detector says a student essay's photo is "70% likely AI." What should a teacher do?

Answer: B. Treat it as one clue and talk with the student

Scores are not proof.

- ✗ A. Ignore the photo: not necessary
- ✗ C. Report the student publicly: harmful
- ✗ D. Give the student a failing grade right away: not proof

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S84-F1-Q4 · question page 1758

AI.MEDIA.07 · Use provenance and detection carefully in practice

A generator adds a visible watermark. What can remove it?

Answer: C. Cropping or editing

Watermarks can be removed.

- ✗ A. Viewing the image: no
- ✗ B. Nothing at all, because watermarks are permanent and cannot be removed by any method: they can be
- ✗ D. Saving the file: not usually

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S84-F1-Q5 · question page 1759

AI.MEDIA.07 · Use provenance and detection carefully in practice

Which approach to a suspicious image is best?

Answer: D. Combine provenance with source and context checks

Several clues together.

- ✗ A. Rely only on how it looks: unreliable
- ✗ B. Rely only on likes: unreliable
- ✗ C. Rely only on a detector score, since detectors are designed to replace every other check: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S84-F2-Q1 · question page 1760

AI.MEDIA.07 · Avoid misusing detector scores

A detector answers YES when it judges an image AI-made. What is a false positive?

Answer: A. Flagging a real photo as AI

Wrongly flagged.

- ✗ B. Correctly passing a real photo: true negative
- ✗ C. Correctly flagging an AI image: true positive
- ✗ D. Missing an AI image: false negative

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S84-F2-Q2 · question page 1761

AI.MEDIA.07 · Avoid misusing detector scores

For the same kind of detector (YES means "AI-made"), what is a false negative?

Answer: C. Missing an AI image

A fake that passes.

- ✗ A. Correctly passing a real photo: true negative
- ✗ B. Correctly finding an AI image: true positive
- ✗ D. Flagging a real photo as AI: false positive

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S84-F2-Q3 · question page 1762

AI.MEDIA.07 · Avoid misusing detector scores

Why is accusing someone based on a score alone harmful?

Answer: D. Scores can be wrong

Errors cause real harm.

- ✗ A. Because detector scores are always exactly 50%, so they never give any information: not true
- ✗ B. Because scores are secret: no
- ✗ C. Because accusations are always illegal: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S84-F2-Q4 · question page 1763

AI.MEDIA.07 · Avoid misusing detector scores

A journalist's real photo is flagged "90% AI." What should an editor do?

Answer: B. Check the source and camera files

Investigate further.

- ✗ A. Retract the photo at once, since 90% is high enough to treat the score as certain proof: not certain
- ✗ C. Fire the journalist: harmful
- ✗ D. Publish the score instead: not helpful

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S84-F2-Q5 · question page 1764

AI.MEDIA.07 · Avoid misusing detector scores

How should provenance and detection be treated?

Answer: A. As one clue among several

One clue of many.

- ✗ B. As useless: they help
- ✗ C. As legal proof: not proof
- ✗ D. As the final answer: not reliable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S85-M-Q1 · question page 1766

AI.CODE.01 · Read a few lines of Python and predict what they do

In this example, what does the program print?

Prepared code example (Python 3). Shown only; nothing runs.

```
price = 4
count = 3
total = price * count
print(total)
```

Answer: C. 12

4 * 3 = 12.

✗ A. 7, because the program adds the price and the count together before printing the total: it multiplies

✗ B. total: it prints the value, not the name

✗ D. 4: that is price

Arithmetic checked by code: yes. Code example run and its output checked: yes. Sources: none (stable)

S85-M-Q2 · question page 1767

AI.CODE.01 · Read a few lines of Python and predict what they do

What does "price = 4" do in Python?

Answer: A. Stores 4 under the name price

"=" stores a value.

✗ B. Deletes price: no

✗ C. Checks whether price is already equal to 4 and prints True or False on the screen: that is ==

✗ D. Prints 4: no print

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S85-M-Q3 · question page 1768

AI.CODE.01 · Read a few lines of Python and predict what they do

If the second line were "count = 5," what would print?

Prepared code example (Python 3). Shown only; nothing runs.

```
price = 4
count = 3
total = price * count
print(total)
```

Answer: D. 20

4 * 5 = 20.

✗ A. 9, because changing the count to 5 adds 5 to the price of 4 instead of multiplying: it multiplies

✗ B. 5: that is count

✗ C. 12: count changed

Arithmetic checked by code: yes. Code example run and its output checked: yes. Sources: none (stable)

S85-M-Q4 · question page 1769

AI.CODE.01 · Read a few lines of Python and predict what they do

In what order does Python run these lines?

Answer: B. Top to bottom

Instructions run in order.

- ✗ A. In random order: no
- ✗ C. Only the last line: no
- ✗ D. Bottom to top: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S85-M-Q5 · question page 1770

AI.CODE.01 · Read a few lines of Python and predict what they do

What does print(...) do?

Answer: C. Shows a value on screen

Displays output.

- ✗ A. Multiplies numbers: that is *
- ✗ B. Stores a value: that is =
- ✗ D. Sends the program: not in Python

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S85-B1-Q1 · question page 1771

AI.CODE.01 · Trace variables on paper

After line 2, what is b?

Answer: A. 5

2 + 3 = 5.

- ✗ B. 3: a + 3
- ✗ C. 10: later
- ✗ D. 2, because b is given the same value as a, and the "+ 3" part is ignored by Python: adds 3

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S85-B1-Q2 · question page 1772

AI.CODE.01 · Trace variables on paper

After line 3, what is a?

Answer: B. 10

a is replaced.

- ✗ A. 12: no
- ✗ C. 2, because once a variable is given a value it can never be changed by a later line: it changes
- ✗ D. 5: that is b

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S85-B1-Q3 · question page 1773

AI.CODE.01 · Trace variables on paper

After line 3, what is b?

Answer: D. 5

b was already calculated.

- ✗ A. 10: that is a
- ✗ B. 2: no
- ✗ C. 13: it does not

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S85-B1-Q4 · question page 1774

AI.CODE.01 · Trace variables on paper

What does line 4 print?

Answer: C. 15

10 + 5 = 15.

- ✗ A. 12: no
- ✗ B. 7, which is the original value of a added to the original value of b from line 2: a changed
- ✗ D. a + b: prints the value

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S85-B1-Q5 · question page 1775

AI.CODE.01 · Trace variables on paper

What is the key tracing habit?

Answer: D. Update a variable only when a line changes it

Cross out and rewrite.

- ✗ A. Track only the last line: no
- ✗ B. Update every variable on every line, even when the line does not mention that variable: no
- ✗ C. Ignore numbers: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S85-F1-Q1 · question page 1776

AI.CODE.01 · Predict the output of short programs

What does this program print?

Prepared code example (Python 3). Shown only; nothing runs.

```
age = 15
if age >= 16:
    print("Can apply")
else:
    print("Too young")
```

Answer: B. Too young

15 >= 16 is false, so else runs.

- ✗ A. 15: not printed
- ✗ C. Nothing: else runs
- ✗ D. Can apply: condition false

Arithmetic checked by code: yes. Code example run and its output checked: yes. Sources: none (stable)

S85-F1-Q2 · question page 1777

AI.CODE.01 · Predict the output of short programs

If the first line were "age = 16," what would print?

Prepared code example (Python 3). Shown only; nothing runs.

```
age = 15
if age >= 16:
    print("Can apply")
else:
    print("Too young")
```

Answer: A. Can apply

16 >= 16 is true.

- ✗ B. 16: not printed
- ✗ C. Both lines: only one branch runs
- ✗ D. Too young: >= includes 16

Arithmetic checked by code: yes. Code example run and its output checked: yes. Sources: none (stable)

S85-F1-Q3 · question page 1778

AI.CODE.01 · Predict the output of short programs

What does this loop print?

Prepared code example (Python 3). Shown only; nothing runs.

```
for n in [1, 2, 3]:
    print(n * 2)
```

Answer: D. 2, 4 and 6 on separate lines

Each n doubled.

- ✗ A. 12: no sum
- ✗ B. 6 only: prints each time
- ✗ C. 1, 2 and 3 on separate lines: it doubles

Arithmetic checked by code: not applicable. Code example run and its output checked: yes.
Sources: none (stable)

S85-F1-Q4 · question page 1779

AI.CODE.01 · Predict the output of short programs

How many times does the loop body run?

Prepared code example (Python 3). Shown only; nothing runs.

```
for n in [1, 2, 3]:
    print(n * 2)
```

Answer: B. 3

Three items in the list.

- ✗ A. 6, because the loop runs once for every number that it prints out on the screen: once per item
- ✗ C. 2: once per item
- ✗ D. 1: once per item

Arithmetic checked by code: not applicable. Code example run and its output checked: yes.
Sources: none (stable)

S85-F1-Q5 · question page 1780

AI.CODE.01 · Predict the output of short programs

In "x = 7" then "x = x + 1," what is x afterward?

Answer: A. 8

7 + 1 = 8.

- ✗ B. 7, because a variable cannot be used on both sides of the same "=" sign in Python: it can
- ✗ C. 71: not text joining
- ✗ D. 1: no

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S85-F2-Q1 · question page 1781

AI.CODE.01 · Read if statements, loops and indentation

What does "if age >= 16:" mean?

Answer: C. If age is at least 16

Greater than or equal.

- ✗ A. Set age to 16: not assignment
- ✗ B. If age is under 16: reversed
- ✗ D. If age is exactly 16: >= includes more

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S85-F2-Q2 · question page 1782

AI.CODE.01 · Read if statements, loops and indentation

What does indentation show in Python?

Answer: A. Which lines belong together

Blocks of code.

- ✗ B. Comments: no
- ✗ C. How important each line is, with the most important lines indented the furthest: not importance
- ✗ D. Nothing; it is decoration: it matters

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S85-F2-Q3 · question page 1783

AI.CODE.01 · Read if statements, loops and indentation

In "for n in [5, 10]: print(n)," what prints?

Answer: C. 5 then 10

Each item in turn.

- ✗ A. Nothing: it prints
- ✗ B. n twice: prints values
- ✗ D. 15: no sum

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S85-F2-Q4 · question page 1784

AI.CODE.01 · Read if statements, loops and indentation

When does an else part run?

Answer: D. When the if condition is false

The other branch.

- ✗ A. Never: it can
- ✗ B. Always, right after the if part has finished, whether the condition was true or false: only when false
- ✗ C. Before the if: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S85-F2-Q5 · question page 1785

AI.CODE.01 · Read if statements, loops and indentation

A loop goes over a list of 4 names. How many times does its body run?

Answer: B. 4

Once per item.

- ✗ A. 8: no
- ✗ C. 0: it runs
- ✗ D. 1, because a loop body runs only once no matter how many items are in the list: once per item

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-M-Q1 · question page 1787

AI.CODE.02 · Explain why code that looks right can be wrong and how to check it

Why can generated code be wrong even when it looks neat?

Answer: A. The model predicts likely code without proving it

Prediction, not proof.

- ✗ B. Because computers cannot run neat code: no
- ✗ C. Because comments stop code from running: no
- ✗ D. Because neat code is always wrong, and only messy code can actually work correctly: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-M-Q2 · question page 1788

AI.CODE.02 · Explain why code that looks right can be wrong and how to check it

In this example, which result is wrong?

Prepared code example (Python 3). Shown only; nothing runs.

```
def average(numbers):
    return sum(numbers) / 2

print(average([4, 6]))
print(average([3, 6, 9]))
```

Answer: D. 9.0 for [3, 6, 9]

The average of 3, 6, 9 is 6.

- ✗ A. Both results: 5.0 is right
- ✗ B. Neither result: one is wrong
- ✗ C. 5.0 for [4, 6]: 5.0 is right

Arithmetic checked by code: yes. Code example run and its output checked: yes. Sources: none (stable)

S86-M-Q3 · question page 1789

AI.CODE.02 · Explain why code that looks right can be wrong and how to check it

What is the bug in the same example?

Prepared code example (Python 3). Shown only; nothing runs.

```
def average(numbers):
    return sum(numbers) / 2

print(average([4, 6]))
print(average([3, 6, 9]))
```

Answer: C. It always divides by 2

It should divide by len(numbers).

- ✗ A. Its name is wrong: not the bug
- ✗ B. It prints too much: not the bug
- ✗ D. It uses sum: sum is fine

Arithmetic checked by code: not applicable. Code example run and its output checked: yes.
Sources: none (stable)

S86-M-Q4 · question page 1790

AI.CODE.02 · Explain why code that looks right can be wrong and how to check it

What does the example show about testing?

Prepared code example (Python 3). Shown only; nothing runs.

```
def average(numbers):
    return sum(numbers) / 2

print(average([4, 6]))
print(average([3, 6, 9]))
```

Answer: B. Passing one test does not prove code right

Test several inputs.

✗ A. Code never needs tests: no

✗ C. Testing is a waste of time, because the first test already showed that the function works: reversed

✗ D. Two tests are always enough: not always

Arithmetic checked by code: not applicable. Code example run and its output checked: yes.
Sources: none (stable)

S86-M-Q5 · question page 1791

AI.CODE.02 · Explain why code that looks right can be wrong and how to check it

What is the best way to check code?

Answer: C. Run it on inputs whose answers you know

Known answers reveal errors.

- ✗ A. Count its lines: not a check
- ✗ B. Ask the tool that wrote it whether it is correct, and accept the answer that it gives: not independent
- ✗ D. Check that it has comments: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-B1-Q1 · question page 1792

AI.CODE.02 · Compare checking code with checking a calculator you built

Why type 2 + 2 into a new calculator?

Answer: B. You already know the answer

Known answers reveal errors.

- ✗ A. To make it faster: no
- ✗ C. To clean the buttons: no
- ✗ D. Because 2 + 2 is the only problem: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-B1-Q2 · question page 1793

AI.CODE.02 · Compare checking code with checking a calculator you built

The calculator gets 2 + 2 right but 5 * 5 wrong. Does it work?

Answer: A. No; one wrong result shows a problem

One failure matters.

- ✗ B. It cannot be known: it can
- ✗ C. Yes, mostly: still broken
- ✗ D. Yes, because getting the first problem right proves that every other answer is right: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-B1-Q3 · question page 1794

AI.CODE.02 · Compare checking code with checking a calculator you built

Nice-looking buttons prove what?

Answer: D. Nothing about correctness

Looks are not tests.

- ✗ A. That every answer the calculator gives will always be correct for any problem: no
- ✗ B. That it is cheap: no
- ✗ C. That it is fast: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-B1-Q4 · question page 1795

AI.CODE.02 · Compare checking code with checking a calculator you built

Generated code is like what in this comparison?

Answer: B. A new calculator to test

Needs testing.

- ✗ A. A broken calculator: not necessarily
- ✗ C. A math book: no
- ✗ D. A calculator that has already been fully tested by thousands of people over many years: not yet tested

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-B1-Q5 · question page 1796

AI.CODE.02 · Compare checking code with checking a calculator you built

What should you test code with?

Answer: D. Inputs whose correct answers you know

Known-answer tests.

- ✗ A. Inputs chosen at random whose correct answers nobody knows, so the test is fair: cannot compare
- ✗ B. Only one input: not enough
- ✗ C. No inputs: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-F1-Q1 · question page 1797

AI.CODE.02 · Check generated code in practice

A tool writes a function to count vowels. Which test helps most?

Answer: C. Run it on "banana" and expect 3

A known answer: a, a, a.

- ✗ A. Check it has a good name: not a test
- ✗ B. Count its lines: not a test
- ✗ D. Read the comment at the top: not a test

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-F1-Q2 · question page 1798

AI.CODE.02 · Check generated code in practice

Generated code calls a function the tool named "sort_smart," which does not exist. What will happen?

Answer: A. An error when it runs

Invented functions fail.

- ✗ B. The computer will create the missing function automatically, based on its name: no
- ✗ C. It will run normally: it does not exist
- ✗ D. It will print the name: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-F1-Q3 · question page 1799

AI.CODE.02 · Check generated code in practice

A function to find the largest number works on [3, 9, 2]. What else should you test?

Answer: B. An empty list and negative numbers

Test boundaries.

- ✗ A. The same list again: no new information
- ✗ C. Nothing else, since working on one list of three numbers shows it will work on every list: not proof
- ✗ D. A list of colors: wrong type

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-F1-Q4 · question page 1800

AI.CODE.02 · Check generated code in practice

Code will calculate students' grades. What extra step is wise?

Answer: A. Have someone experienced review it

Higher stakes, more review.

- ✗ B. Make it shorter: not a check
- ✗ C. Delete the tests: no
- ✗ D. Use it immediately for every student: affects people

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-F1-Q5 · question page 1801

AI.CODE.02 · Check generated code in practice

Which shows that code really works?

Answer: C. Correct results on several known-answer tests

Behavior is evidence.

- ✗ A. Neat indentation: not proof
- ✗ B. A long, confident explanation from the tool describing exactly what the code is doing: not proof
- ✗ D. Many comments: not proof

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-F2-Q1 · question page 1802

AI.CODE.02 · Trust behavior over descriptions

A tool explains its code "removes duplicates," but running it leaves duplicates. Which should you trust?

Answer: D. What the code actually does when run

Run it.

- ✗ A. Neither: behavior is evidence
- ✗ B. The explanation, since the tool knows its own code better than any test that you could run: behavior wins
- ✗ C. The comments: can be wrong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-F2-Q2 · question page 1803

AI.CODE.02 · Trust behavior over descriptions

A comment says "# adds tax," but the line subtracts. What does this show?

Answer: A. Comments can be wrong

Read the code itself.

- ✗ B. That the subtraction must be correct, because comments are always checked before code runs: no
- ✗ C. That the code will not run: it runs
- ✗ D. That tax is negative: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-F2-Q3 · question page 1804

AI.CODE.02 · Trust behavior over descriptions

Which inputs test the boundaries of a function that handles ages?

Answer: D. 0, a very large age, and a missing age

Smallest, largest, missing.

- ✗ A. Only 25: one value
- ✗ B. Only names: wrong type
- ✗ C. Three typical adult ages, such as 30, 35 and 40, which cover the most common situations: not boundaries

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-F2-Q4 · question page 1805

AI.CODE.02 · Trust behavior over descriptions

Why do confident explanations make wrong code more convincing?

Answer: B. They make you less likely to test it

Tone discourages checking.

- ✗ A. Because they contain numbers: not the point
- ✗ C. Because confident explanations actually change the code so that it works correctly: no
- ✗ D. Because they are longer: not the point

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-F2-Q5 · question page 1806

AI.CODE.02 · Trust behavior over descriptions

Code will handle customer payments. What should happen before use?

Answer: C. Experienced review and testing

Money needs review.

- ✗ A. Only a spell check: not enough
- ✗ B. Only a quick glance: not enough
- ✗ D. Nothing, since payment code written by a tool is always reviewed by a bank automatically: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-M-Q1 · question page 1808

AI.CODE.03 · State requirements clearly, including edge cases

What is a requirement?

Answer: C. A checkable statement of what code must do

Clear, checkable.

- ✗ A. A guess about what users want: not checkable
- ✗ B. The program's file name: no
- ✗ D. A list of the programming languages: not a requirement

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-M-Q2 · question page 1809

AI.CODE.03 · State requirements clearly, including edge cases

Which is an edge case for a test-score average?

Answer: A. An empty list of scores

Empty input often breaks code.

- ✗ B. A list of ten scores: typical
- ✗ C. Scores of 80 and 85: typical
- ✗ D. A typical list: typical case

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-M-Q3 · question page 1810

AI.CODE.03 · State requirements clearly, including edge cases

In this example, which input would break the function?

Prepared code example (Python 3). Shown only; nothing runs.

```
def first_letter(name):
    return name[0]

print(first_letter("Maya"))
```

Answer: B. An empty name ""

name[0] fails when there is no first letter.

✗ A. "Zoe": works

✗ C. "Al": works

✗ D. "Maya," since names with four letters are too long for this function to handle correctly: it works

Arithmetic checked by code: not applicable. Code example run and its output checked: yes.
Sources: none (stable)

S87-M-Q4 · question page 1811

AI.CODE.03 · State requirements clearly, including edge cases

Which is a boundary value for scores from 0 to 100?

Answer: D. 100

Exactly at the limit.

- ✗ A. 72: middle
- ✗ B. 85: middle
- ✗ C. 57: not a boundary

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-M-Q5 · question page 1812

AI.CODE.03 · State requirements clearly, including edge cases

Which request includes an edge-case rule?

Answer: A. "If the list is empty, return 0 and do not crash."

Says what to do in an unusual case.

- ✗ B. "Use good names.": no edge case
- ✗ C. "Make it fast.": no edge case
- ✗ D. "Write the code for the average quickly, and make it look clean and professional.": no edge case

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-B1-Q1 · question page 1813

AI.CODE.03 · Compare edge cases with testing a door lock

Why does a locksmith try the wrong key?

Answer: B. To see how the lock handles unusual situations

Test unusual cases.

- ✗ A. To save time: no
- ✗ C. To damage the lock: no
- ✗ D. Because the right key never works on any new lock: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-B1-Q2 · question page 1814

AI.CODE.03 · Compare edge cases with testing a door lock

"No key at all" is like which code input?

Answer: C. An empty input

Nothing provided.

- ✗ A. A very large number: different
- ✗ B. A correct name: no
- ✗ D. A typical input: not unusual

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-B1-Q3 · question page 1815

AI.CODE.03 · Compare edge cases with testing a door lock

Freezing weather is like what in code?

Answer: D. Unusual conditions

Stress conditions.

- ✗ A. A comment: no
- ✗ B. A variable name: no
- ✗ C. Normal use: not normal

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-B1-Q4 · question page 1816

AI.CODE.03 · Compare edge cases with testing a door lock

Where do locks and code most often fail?

Answer: A. In unusual situations

Edge cases.

- ✗ B. Never: they fail
- ✗ C. Only on weekends: no
- ✗ D. In the most common situation, the one that the designer tested most carefully: least likely

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-B1-Q5 · question page 1817

AI.CODE.03 · Compare edge cases with testing a door lock

What is the lesson for code?

Answer: D. Test unusual inputs, not just the normal one

Edge cases matter.

- ✗ A. Only test long inputs: too narrow
- ✗ B. Never test code: no
- ✗ C. Test only the normal input once, because unusual inputs never happen in real life: they happen

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-F1-Q1 · question page 1818

AI.CODE.03 · Write requirements with edge cases in practice

Ben asks: "Write code to split a bill." What key detail is missing?

Answer: C. How many people and how to round

Inputs and rounding rules.

- ✗ A. The restaurant's address: irrelevant
- ✗ B. The programmer's name: irrelevant
- ✗ D. Which color to use for the numbers: not key

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-F1-Q2 · question page 1819

AI.CODE.03 · Write requirements with edge cases in practice

A function divides a total by the number of people. Which edge case matters most?

Answer: B. Zero people

Dividing by zero crashes.

- ✗ A. Three people: typical
- ✗ C. Two people: typical
- ✗ D. Four people: typical

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-F1-Q3 · question page 1820

AI.CODE.03 · Write requirements with edge cases in practice

Which requirement is clearest?

Answer: D. "Return the total price including 8% tax, rounded to cents."

Specific and checkable.

- ✗ A. "Do prices.": vague
- ✗ B. "Handle tax.": vague
- ✗ C. "Make the price thing work properly with tax and everything else.": vague

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-F1-Q4 · question page 1821

AI.CODE.03 · Write requirements with edge cases in practice

A form asks for age. Which input is a wrong-type edge case?

Answer: B. "twelve" typed as a word

Text where a number is expected.

- ✗ A. 15: typical
- ✗ C. 30: typical
- ✗ D. 12, typed as a whole number: typical

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-F1-Q5 · question page 1822

AI.CODE.03 · Write requirements with edge cases in practice

A score must be from 0 to 100. Which two inputs test the boundaries?

Answer: C. 0 and 100

Exactly at the limits.

- ✗ A. 60 and 61: not boundaries
- ✗ B. 20 and 30: not boundaries
- ✗ D. 50 and 75: not boundaries

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-F2-Q1 · question page 1823

AI.CODE.03 · Fill gaps in requirements before coding

"Sort the names" leaves which question open?

Answer: A. Alphabetical or by length?

Unstated rules get guessed.

- ✗ B. Which font to show: irrelevant
- ✗ C. Which keyboard to use: irrelevant
- ✗ D. Nothing: not settled

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-F2-Q2 · question page 1824

AI.CODE.03 · Fill gaps in requirements before coding

Why write input examples with expected outputs?

Answer: B. They clarify the requirement and become tests

Examples double as tests.

- ✗ A. Because tools need exactly ten examples: no
- ✗ C. To make the request longer: not the point
- ✗ D. Because examples make the code run faster: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-F2-Q3 · question page 1825

AI.CODE.03 · Fill gaps in requirements before coding

You are unsure whether "Ana" and "ana" count as the same name. What should you do?

Answer: D. Decide the rule before coding

Choose the rule first.

- ✗ A. Delete one of the names: not a rule
- ✗ B. Ignore the question: leads to bugs
- ✗ C. Let the tool decide quietly: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-F2-Q4 · question page 1826

AI.CODE.03 · Fill gaps in requirements before coding

What happens when a requirement is unstated?

Answer: A. The tool fills it with a guess

Guesses may not fit.

- ✗ B. The tool always asks you a question first and never guesses anything on its own: not always
- ✗ C. The code refuses to run: not usually
- ✗ D. Nothing at all: something is chosen

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-F2-Q5 · question page 1827

AI.CODE.03 · Fill gaps in requirements before coding

Which example helps clarify "remove extra spaces"?

Answer: C. " hi there " becomes "hi there"

Concrete input and output.

- ✗ A. "hello" stays "hello": no spaces involved
- ✗ B. "Spaces are bad": not an example
- ✗ D. "Make the text look tidier and cleaner": vague

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-M-Q1 · question page 1829

AI.CODE.04 · Choose a test whose expected result would reveal a defect

What does a test compare?

Answer: D. The code's result with an expected result you worked out

Actual versus expected.

- ✗ A. The code's speed with the internet speed: no
- ✗ B. Two programmers' opinions: not a test
- ✗ C. The code's length with the length of another program: not a test

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-M-Q2 · question page 1830

AI.CODE.04 · Choose a test whose expected result would reveal a defect

In this example, which test reveals the bug?

Prepared code example (Python 3). Shown only; nothing runs.

```
def is_adult(age):
    return age > 18

print(is_adult(20))
print(is_adult(18))
```

Answer: A. is_adult(18)

18 should be True but gives False.

- ✗ B. is_adult(50): passes
- ✗ C. is_adult(30): passes
- ✗ D. is_adult(20): it passes

Arithmetic checked by code: not applicable. Code example run and its output checked: yes.
Sources: none (stable)

S88-M-Q3 · question page 1831

AI.CODE.04 · Choose a test whose expected result would reveal a defect

What is the bug in the same example?

Prepared code example (Python 3). Shown only; nothing runs.

```
def is_adult(age):
    return age > 18

print(is_adult(20))
print(is_adult(18))
```

Answer: B. It uses > instead of >=

18 or older needs >=.

- ✗ A. It prints too much: not the bug
- ✗ C. It has no name: it does
- ✗ D. It uses the word "age,": allowed

Arithmetic checked by code: not applicable. Code example run and its output checked: yes.
Sources: none (stable)

S88-M-Q4 · question page 1832

AI.CODE.04 · Choose a test whose expected result would reveal a defect

What makes a test good?

Answer: C. It would fail if a particular mistake were present

Targeted at a mistake.

- ✗ A. It is very long: not the point
- ✗ B. It uses random inputs: not targeted
- ✗ D. It always passes, which shows that the code is working correctly every single time: not useful

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-M-Q5 · question page 1833

AI.CODE.04 · Choose a test whose expected result would reveal a defect

To catch a mistake at a limit, what should you test?

Answer: B. The limit itself

Boundaries expose errors.

- ✗ A. Nothing: must test
- ✗ C. Only typical values: misses it
- ✗ D. A value far away: misses it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-B1-Q1 · question page 1834

AI.CODE.04 · Pick the input that exposes a suspected mistake

Which input exposes > 50 versus >= 50?

Answer: C. 50

Only 50 differs.

- ✗ A. 30: same either way
- ✗ B. 100: same either way
- ✗ D. 80: same either way

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-B1-Q2 · question page 1835

AI.CODE.04 · Pick the input that exposes a suspected mistake

For input 50, what should the test expect?

Answer: A. Free shipping

"50 or more" includes 50.

- ✗ B. An error: no
- ✗ C. Half-price shipping: no
- ✗ D. Not free: rule says or more

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-B1-Q3 · question page 1836

AI.CODE.04 · Pick the input that exposes a suspected mistake

Why do 30 and 80 not help?

Answer: D. Both rules give the same answer for them

No difference to detect.

- ✗ A. They crash the code: no
- ✗ B. They are too large: no
- ✗ C. Because 30 and 80 are not real prices: they are fine

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-B1-Q4 · question page 1837

AI.CODE.04 · Pick the input that exposes a suspected mistake

What question drives the method?

Answer: C. Which input gives a different answer if the mistake exists?

Target the difference.

- ✗ A. Which input is easiest to type?: not the question
- ✗ B. Which input is the most common?: not the question
- ✗ D. Which input is the largest number that the program is able to handle without crashing?: not the question

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-B1-Q5 · question page 1838

AI.CODE.04 · Pick the input that exposes a suspected mistake

Rule: "under 13 pays child price." Suspected mistake: <= 13. Which input exposes it?

Answer: D. 13

Only 13 differs.

- ✗ A. 12: same either way
- ✗ B. 5, because young children are the main customers who pay the child price at the door: same either way
- ✗ C. 20: same either way

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-F1-Q1 · question page 1839

AI.CODE.04 · Design tests that catch mistakes

A function should give a discount to orders of 100 dollars or more. Which test is most useful?

Answer: A. An order of exactly 100, expecting a discount

Test the boundary.

- ✗ B. An order of 500, which is so large that it should definitely get the discount in any case: misses the limit
- ✗ C. No test: must test
- ✗ D. An order of 20, expecting no discount: misses the limit

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-F1-Q2 · question page 1840

AI.CODE.04 · Design tests that catch mistakes

A function should count words. Which test could reveal a mistake with extra spaces?

Answer: B. "hi there" (two spaces), expecting 2

Target the suspected mistake.

- ✗ A. "hello," a single word with no spaces at all, which is the simplest possible test: no spaces
- ✗ C. An empty test: not targeted
- ✗ D. "one two" with one space: typical

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-F1-Q3 · question page 1841

AI.CODE.04 · Design tests that catch mistakes

Ana writes a test expecting the average of [2, 4] to be 3. The code returns 3. What does this show?

Answer: D. The code passes this test

One pass, not full proof.

- ✗ A. That the test is wrong: it is right
- ✗ B. Nothing at all: it shows a pass
- ✗ C. That the code is correct: one test

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S88-F1-Q4 · question page 1842

AI.CODE.04 · Design tests that catch mistakes

A function should reject passwords shorter than 8 characters. Which input tests the boundary?

Answer: B. A password of exactly 7 characters, expecting rejection

Just below the limit.

- ✗ A. A password of 30 characters, since longer passwords are the most important ones to check: far from limit
- ✗ C. A password of 15 characters: far from limit
- ✗ D. No password: different case

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-F1-Q5 · question page 1843

AI.CODE.04 · Design tests that catch mistakes

Which expected result should a test use?

Answer: C. One you worked out yourself

Independent expected answers.

- ✗ A. The tool's guess: not independent
- ✗ B. Whatever the code printed the first time it ran, since the code is the best source of answers: circular
- ✗ D. A random number: not a check

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-F2-Q1 · question page 1844

AI.CODE.04 · Spot weak tests

A test's expected answer was copied from the code's output. What is the problem?

Answer: A. It only checks the code agrees with itself

Not independent.

- ✗ B. The test uses numbers: not the problem
- ✗ C. Nothing, because the code's own output is always the most reliable expected answer: circular
- ✗ D. The test is too long: not the problem

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-F2-Q2 · question page 1845

AI.CODE.04 · Spot weak tests

A test checks only that the result "is a number." Why is it weak?

Answer: C. It passes even when the number is wrong

Checks too little.

- ✗ A. Because numbers are not results: they are
- ✗ B. Because it is too strict: too loose
- ✗ D. Because numbers can never be checked by tests: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-F2-Q3 · question page 1846

AI.CODE.04 · Spot weak tests

All tests use typical inputs. What might they miss?

Answer: B. Bugs at boundaries and special cases

Untested paths.

- ✗ A. Spelling mistakes in comments: not the point
- ✗ C. Nothing, since typical inputs exercise every single line of code in every program: not always
- ✗ D. The program's file name: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-F2-Q4 · question page 1847

AI.CODE.04 · Spot weak tests

Which test is strongest for a function that returns the largest of three numbers?

Answer: D. Largest in last place: (1, 2, 9) expecting 9

Specific expected answer, tricky position.

- ✗ A. No test: must test
- ✗ B. (5, 5, 5) expecting a result: checks too little
- ✗ C. (9, 2, 1) expecting a number of some kind: checks too little

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-F2-Q5 · question page 1848

AI.CODE.04 · Spot weak tests

Where should strong tests focus?

Answer: A. Boundaries and special cases, with independent answers

Targeted and independent.

- ✗ B. Only the program name: no
- ✗ C. Random inputs with no expected answers: cannot compare
- ✗ D. The most common input only, run many times in a row: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-M-Q1 · question page 1850

AI.CODE.05 · Use an error message to locate a problem

Where should you start reading a traceback?

Answer: A. At the bottom

The last line names the error.

- ✗ B. In the middle: no
- ✗ C. Nowhere; ignore it: it helps
- ✗ D. At the very top: it does not

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-M-Q2 · question page 1851

AI.CODE.05 · Use an error message to locate a problem

In this example, what type of error is shown?

Prepared code example (Python 3) and its error message, simplified. Shown only; nothing runs.

```
scores = [90, 85, 77]
print(scores[3])

Traceback (most recent call last):
File "grades.py", line 2, in <module>
print(scores[3])
IndexError: list index out of range
```

Answer: B. IndexError

Position 3 does not exist.

- ✗ A. TypeError: no
- ✗ C. ValueError: no
- ✗ D. NameError: it was

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-M-Q3 · question page 1852

AI.CODE.05 · Use an error message to locate a problem

In the same example, why is position 3 out of range?

Prepared code example (Python 3) and its error message, simplified. Shown only; nothing runs.

```
scores = [90, 85, 77]
print(scores[3])

Traceback (most recent call last):
File "grades.py", line 2, in <module>
print(scores[3])
IndexError: list index out of range
```

Answer: C. Python counts positions from 0, so they are 0 to 2

Three items: positions 0, 1, 2.

- ✗ A. Because 77 is too large: no
- ✗ B. Because print cannot show lists: no
- ✗ D. Because lists in Python can only hold two items: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-M-Q4 · question page 1853

AI.CODE.05 · Use an error message to locate a problem

In this example, what caused the NameError?

Prepared code example (Python 3) and its error message, simplified. Shown only; nothing runs.

```
total = 50
print(totl)

Traceback (most recent call last):
File "shop.py", line 2, in <module>
print(totl)
NameError: name 'totl' is not defined
```

Answer: D. A typo: totl instead of total

A name never defined.

- ✗ A. print is misspelled: it is not
- ✗ B. The file name is wrong: no
- ✗ C. The number 50 is too large: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-M-Q5 · question page 1854

AI.CODE.05 · Use an error message to locate a problem

When asking a tool to fix an error, what should you include?

Answer: B. The relevant error message, with secrets removed

Give the relevant message and code, after removing keys or personal details.

- ✗ A. Only the words "it is broken," since the tool can work out the rest by itself: too little
- ✗ C. Nothing: too little
- ✗ D. Only the file name: too little

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-B1-Q1 · question page 1855

AI.CODE.05 · Read the last two lines of a traceback

In the example, which file has the error?

Answer: C. pay.py

Named in the File line.

- ✗ A. line 7: that is the line
- ✗ B. ZeroDivisionError, which is the name of the file where the problem happened: that is the type
- ✗ D. rate: that is a variable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-B1-Q2 · question page 1856

AI.CODE.05 · Read the last two lines of a traceback

Which line number?

Answer: A. 7

From the File line.

- ✗ B. 0, because the error message mentions division by zero in the last line: not a line number
- ✗ C. 1: no
- ✗ D. 2: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-B1-Q3 · question page 1857

AI.CODE.05 · Read the last two lines of a traceback

What is the error type?

Answer: D. ZeroDivisionError

The last line.

- ✗ A. hours: no
- ✗ B. line 7: no
- ✗ C. pay.py, which is listed just above the line of code that caused the problem: that is the file

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-B1-Q4 · question page 1858

AI.CODE.05 · Read the last two lines of a traceback

What is the problem?

Answer: B. Dividing by zero

Division by zero.

- ✗ A. Adding text to numbers: no
- ✗ C. A missing file: no
- ✗ D. A misspelled variable name: not this one

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-B1-Q5 · question page 1859

AI.CODE.05 · Read the last two lines of a traceback

Which two lines hold most of the information?

Answer: A. The last two

Read from the bottom.

- ✗ B. None: they help
- ✗ C. The middle two: no
- ✗ D. The first two: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-F1-Q1 · question page 1860

AI.CODE.05 · Locate problems from error messages

In this example, what is the problem?

Prepared code example (Python 3) and its error message, simplified. Shown only; nothing runs.

```
age = 15
print("Age: " + age)

Traceback (most recent call last):
File "card.py", line 2, in <module>
print("Age: " + age)
TypeError: can only concatenate str (not "int") to str
```

Answer: D. Adding text to a number

TypeError: str plus int.

- ✗ A. The variable name "age" is not allowed in Python, so the program cannot run at all: allowed
- ✗ B. A missing file: no
- ✗ C. The number 15 is too large: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-F1-Q2 · question page 1861

AI.CODE.05 · Locate problems from error messages

In the same example, which line has the problem?

Prepared code example (Python 3) and its error message, simplified. Shown only; nothing runs.

```
age = 15
print("Age: " + age)

Traceback (most recent call last):
File "card.py", line 2, in <module>
print("Age: " + age)
TypeError: can only concatenate str (not "int") to str
```

Answer: C. Line 2

Named in the traceback.

- ✗ A. Line 3: there is none
- ✗ B. Line 15: that is the age
- ✗ D. Line 1: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-F1-Q3 · question page 1862

AI.CODE.05 · Locate problems from error messages

int("ten") gives "ValueError: invalid literal for int()." What does it mean?

Answer: D. The text is not a number Python can convert

Right type, wrong content.

- ✗ A. The list is too short: IndexError
- ✗ B. The file is missing: no
- ✗ C. The program forgot to define a variable called "ten" before using it on this line: that is NameError

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-F1-Q4 · question page 1863

AI.CODE.05 · Locate problems from error messages

A NameError names "prnt." What is the likely cause?

Answer: C. A typo for print

Misspelled name.

- ✗ A. Text plus a number: no
- ✗ B. Dividing by zero: no
- ✗ D. A missing list item: IndexError

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-F1-Q5 · question page 1864

AI.CODE.05 · Locate problems from error messages

Before asking a tool for a fix, what is a good first step?

Answer: A. Find the line and error type yourself

Locate the problem.

- ✗ B. Restart the computer: not a fix
- ✗ C. Ignore the message: it helps
- ✗ D. Delete the whole program and start again: unnecessary

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-F2-Q1 · question page 1865

AI.CODE.05 · Trace errors to their cause and confirm fixes

An error appears on line 20, but the wrong value was created on line 5. Where is the cause?

Answer: B. Line 5

Errors can surface later.

- ✗ A. Nowhere: there is a cause
- ✗ C. Line 1: not stated
- ✗ D. Line 20: not always

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-F2-Q2 · question page 1866

AI.CODE.05 · Trace errors to their cause and confirm fixes

A tool "fixes" an error by making the program ignore all errors silently. What is wrong?

Answer: D. It hides the problem

Address the cause.

- ✗ A. It uses Python: not the issue
- ✗ B. It is too short: not the issue
- ✗ C. Nothing: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-F2-Q3 · question page 1867

AI.CODE.05 · Trace errors to their cause and confirm fixes

After applying a fix, what should you do?

Answer: A. Rerun the original failing case

Confirm the fix.

- ✗ B. Delete the test: no
- ✗ C. Ask for another fix: test first
- ✗ D. Move on without testing, since the tool already confirmed the fix would definitely work: confirm

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-F2-Q4 · question page 1868

AI.CODE.05 · Trace errors to their cause and confirm fixes

The error is gone, but the output is now wrong. Is it fixed?

Answer: C. No; the result must also be right

No error is not the same as correct.

- ✗ A. Yes, if it is fast: no
- ✗ B. Yes, because removing an error message is the same thing as fixing the problem: not the same
- ✗ D. It cannot be known: it can

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-F2-Q5 · question page 1869

AI.CODE.05 · Trace errors to their cause and confirm fixes

Which fix addresses the cause of IndexError from scores[3] on a 3-item list?

Answer: B. Use a valid position, such as scores[2]

Fix the position.

- ✗ A. Hide the error message so that the program keeps running and prints nothing at all: hides it
- ✗ C. Rename the list: not the cause
- ✗ D. Add more print lines: not a fix

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-M-Q1 · question page 1871

AI.CODE.06 · Use explanations as a starting point and review code against requirements

How should you treat a tool's explanation of code?

Answer: C. As a starting point to check

Helpful, but verify.

- ✗ A. As useless: it helps
- ✗ B. As a replacement for running it: no
- ✗ D. As proof of what the code does: not proof

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-M-Q2 · question page 1872

AI.CODE.06 · Use explanations as a starting point and review code against requirements

In this example, what does the program print?

Prepared code example (Python 3). Shown only; nothing runs.

```
# Return the largest number in the list
def biggest(nums):
    return min(nums)

print(biggest([4, 9, 2]))
```

Answer: A. 2

min() returns the smallest.

✗ B. 4: no

✗ C. 9, because the comment says the function returns the largest number in the list: comment is wrong

✗ D. 15: no sum

Arithmetic checked by code: not applicable. Code example run and its output checked: yes.
Sources: none (stable)

S90-M-Q3 · question page 1873

AI.CODE.06 · Use explanations as a starting point and review code against requirements

In the same example, what is wrong?

Prepared code example (Python 3). Shown only; nothing runs.

```
# Return the largest number in the list
def biggest(nums):
    return min(nums)

print(biggest([4, 9, 2]))
```

Answer: D. The code uses min() but the requirement is the largest

Code does not match the requirement.

- ✗ A. The comment uses the word "largest," which is not allowed in Python comments: allowed
- ✗ B. Nothing: it is wrong
- ✗ C. The list is too short: no

Arithmetic checked by code: not applicable. Code example run and its output checked: yes. Sources: none (stable)

S90-M-Q4 · question page 1874

AI.CODE.06 · Use explanations as a starting point and review code against requirements

What does a code review check?

Answer: B. Each line against the requirements

Line by line, against requirements.

- ✗ A. Only the file name: no
- ✗ C. Only the indentation: no
- ✗ D. Only that the code has comments: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-M-Q5 · question page 1875

AI.CODE.06 · Use explanations as a starting point and review code against requirements

What can be wrong in code besides the statements?

Answer: A. Comments, names and explanations

All can mislead.

- ✗ B. Only the spaces: no
- ✗ C. Only the colors: no
- ✗ D. Nothing; only the statements: others can mislead

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-B1-Q1 · question page 1876

AI.CODE.06 · Compare reviewing code with proofreading a recipe

Why not stop at the recipe title?

Answer: B. It says what it should be, not what the lines do

Check the lines.

- ✗ A. Because cakes have no titles: no
- ✗ C. Because titles are too short: not the point
- ✗ D. Because titles of recipes are always wrong: too strong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-B1-Q2 · question page 1877

AI.CODE.06 · Compare reviewing code with proofreading a recipe

The recipe title is like what in code?

Answer: C. A comment

Intent, not behavior.

- ✗ A. An error message: no
- ✗ B. A variable: no
- ✗ D. A test that runs the code on several inputs and compares the results with answers: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-B1-Q3 · question page 1878

AI.CODE.06 · Compare reviewing code with proofreading a recipe

Checking "cup" versus "tablespoon" is like what?

Answer: D. Checking each line's details

Line-by-line review.

- ✗ A. Reading the title of the recipe twice to make sure that the cake is called the right thing: not detail
- ✗ B. Choosing a font: no
- ✗ C. Counting pages: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-B1-Q4 · question page 1879

AI.CODE.06 · Compare reviewing code with proofreading a recipe

Noticing missing eggs is like what in a code review?

Answer: A. Finding something missing

Look for what is not there.

- ✗ B. Renaming a variable: no
- ✗ C. Running the code: no
- ✗ D. Finding a misspelled comment: not missing

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-B1-Q5 · question page 1880

AI.CODE.06 · Compare reviewing code with proofreading a recipe

What decides whether the cake will work?

Answer: D. The actual lines

Behavior comes from statements.

- ✗ A. The title of the recipe, which tells the baker exactly how the cake will turn out: not the title
- ✗ B. The photo: no
- ✗ C. The page color: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-F1-Q1 · question page 1881

AI.CODE.06 · Review code against requirements in practice

Requirement: "Return the number of students who passed (score 60 or more)." The code counts scores above 60. What is the issue?

Answer: C. It misses students who scored exactly 60

Above 60 excludes 60.

- ✗ A. It uses numbers: no
- ✗ B. It counts too many: too few
- ✗ D. Nothing, since a score of exactly 60 is so rare: requirement includes it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-F1-Q2 · question page 1882

AI.CODE.06 · Review code against requirements in practice

A tool explains code as "sorts names alphabetically," but output shows them by length. What should you trust?

Answer: B. The output

Behavior is evidence.

- ✗ A. Neither: output is evidence
- ✗ C. The comment: can be wrong
- ✗ D. The explanation: behavior wins

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-F1-Q3 · question page 1883

AI.CODE.06 · Review code against requirements in practice

During review, you find a line whose purpose you cannot tell. What should you do?

Answer: A. Ask about it and test it before approving

Unclear code is a finding.

- ✗ B. Approve it anyway, since lines you do not understand are usually the most clever parts of code: risky
- ✗ C. Delete it without checking: may break things
- ✗ D. Ignore it: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-F1-Q4 · question page 1884

AI.CODE.06 · Review code against requirements in practice

Which review question asks about missing parts?

Answer: D. "Is anything the requirement needs not here?"

Look for omissions.

- ✗ A. "Is the font readable?": no
- ✗ B. "Is it short?": no
- ✗ C. "Is the variable name long enough to describe exactly what the variable is used for?": naming

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-F1-Q5 · question page 1885

AI.CODE.06 · Review code against requirements in practice

Which is the best use of a code explanation?

Answer: C. To learn what to check, then verify by running

Starting point.

- ✗ A. To decide the requirement: requirements come first
- ✗ B. To skip reading the code: no
- ✗ D. To replace testing entirely, since a clear explanation proves that the code is correct: not proof

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-F2-Q1 · question page 1886

AI.CODE.06 · Review for what is missing and keep changes reviewable

Code divides by the number of items but never checks for zero items. What kind of finding is this?

Answer: B. A missing check

Something absent.

- ✗ A. A comment problem: no
- ✗ C. No problem: it can crash
- ✗ D. A naming problem: not naming

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-F2-Q2 · question page 1887

AI.CODE.06 · Review for what is missing and keep changes reviewable

A tool proposes changing 400 lines at once. What helps review?

Answer: A. Asking for smaller changes

Small changes are reviewable.

- ✗ B. Reading only the first line: not enough
- ✗ C. Skipping review: risky
- ✗ D. Approving all 400 lines quickly: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-F2-Q3 · question page 1888

AI.CODE.06 · Review for what is missing and keep changes reviewable

You cannot understand part of a generated function. What is the right response?

Answer: C. Ask for a simpler version or verify with small examples

Understanding is part of review.

- ✗ A. Pretend you understand: no
- ✗ B. Remove the whole program: excessive
- ✗ D. Approve it, because code that is hard to understand is usually more efficient and better: not a reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-F2-Q4 · question page 1889

AI.CODE.06 · Review for what is missing and keep changes reviewable

A requirement says "log every failed login," but the code logs only some. What is this?

Answer: B. A skipped requirement

Missing behavior.

- ✗ A. A faster design: not the point
- ✗ C. A typo: no
- ✗ D. A style choice: not optional

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-F2-Q5 · question page 1890

AI.CODE.06 · Review for what is missing and keep changes reviewable

Which review finding is most serious?

Answer: D. An unhandled error that can crash the program

Crashes matter.

- ✗ A. A blank line: minor
- ✗ B. A long variable name: minor
- ✗ C. A comment that uses slightly informal language: minor

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-M-Q1 · question page 1892

AI.CODE.07 · Explain commits, diffs and reverting

What is a commit?

Answer: B. A saved snapshot of code with a message

Snapshot plus message.

- ✗ A. A promise from the programmer to finish the project by a certain date in the future: not the meaning
- ✗ C. An error message: no
- ✗ D. A deleted file: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-M-Q2 · question page 1893

AI.CODE.07 · Explain commits, diffs and reverting

In a diff, what does a line starting with "+" mean?

Answer: A. It was added

Plus means added.

- ✗ B. It is a comment: no
- ✗ C. It has an error: no
- ✗ D. It was removed: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-M-Q3 · question page 1894

AI.CODE.07 · Explain commits, diffs and reverting

In this example, what changed?

Prepared example (a diff in a fictional project). Shown only; nothing runs.

```
- total = price * count
+ total = price + count
```

Answer: C. Multiplication became addition

"*" removed, "+" added.

- ✗ A. The variable was renamed: no
- ✗ B. Nothing changed: a line changed
- ✗ D. Addition became multiplication: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-M-Q4 · question page 1895

AI.CODE.07 · Explain commits, diffs and reverting

What does reverting do?

Answer: D. Undoes a commit with the opposite change

Undo a change.

- ✗ A. Copies the project: no
- ✗ B. Runs the tests: no
- ✗ C. Deletes the whole project and its history: not reverting

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-M-Q5 · question page 1896

AI.CODE.07 · Explain commits, diffs and reverting

Why commit before accepting generated changes?

Answer: B. To see what changed and undo it if needed

A safe restore point.

- ✗ A. To hide changes: no
- ✗ C. To make the code run faster: no
- ✗ D. Because tools refuse to change code: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-B1-Q1 · question page 1897

AI.CODE.07 · Compare version control with save points in a video game

A game save is like what?

Answer: C. A commit

A snapshot.

- ✗ A. A test: no
- ✗ B. A bug: no
- ✗ D. A diff: not a diff

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-B1-Q2 · question page 1898

AI.CODE.07 · Compare version control with save points in a video game

Comparing two saves to see what changed is like what?

Answer: A. A diff

What changed.

- ✗ B. A new game: no
- ✗ C. A revert: no
- ✗ D. A commit message: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-B1-Q3 · question page 1899

AI.CODE.07 · Compare version control with save points in a video game

Loading an earlier save is like what?

Answer: A. Restoring files from an earlier commit

Loading a save puts things back the way they were saved, like restoring files from a commit.

- ✗ B. Buying a game: no
- ✗ C. Committing a brand-new change that adds a new level to the game for every player: no
- ✗ D. Deleting the game: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-B1-Q4 · question page 1900

AI.CODE.07 · Compare version control with save points in a video game

Saving before a friend plays is like what?

Answer: A. Committing before accepting AI changes

A restore point first.

- ✗ B. Deleting every save file so that your friend can start the whole game from the beginning: opposite
- ✗ C. Changing the controls: no
- ✗ D. Turning off the game: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-B1-Q5 · question page 1901

AI.CODE.07 · Compare version control with save points in a video game

Why save before a hard level?

Answer: C. To return if things go badly

Safety net.

- ✗ A. Because saving makes the next level easier to beat on the very first try: no
- ✗ B. To end the game: no
- ✗ D. To change the music: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-F1-Q1 · question page 1902

AI.CODE.07 · Use commits and diffs with AI changes

Ravi committed his working project, then let a tool edit it. Tests now fail, and he has not committed the edit. What can he do?

Answer: C. Restore the files from his last commit

His last commit is a saved good version; restoring returns the files to it.

- ✗ A. Nothing, because a tool's edits can never be undone: the commit lets him restore
- ✗ B. Delete the project: unnecessary
- ✗ D. Ignore the tests: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-F1-Q2 · question page 1903

AI.CODE.07 · Use commits and diffs with AI changes

A diff shows "- if age >= 18:" and "+ if age > 18:". What changed?

Answer: B. The rule now excludes exactly 18

Boundary changed.

- ✗ A. Nothing important: it matters
- ✗ C. The variable name: no
- ✗ D. The rule now includes more ages: it excludes 18

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-F1-Q3 · question page 1904

AI.CODE.07 · Use commits and diffs with AI changes

Which commit message is most useful?

Answer: C. "Fix average to handle an empty list"

What and why.

- ✗ A. "asdf": meaningless
- ✗ B. "stuff": vague
- ✗ D. "Changes," which tells everyone that something in the project was changed at that time: vague

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-F1-Q4 · question page 1905

AI.CODE.07 · Use commits and diffs with AI changes

Why read the diff after a tool edits code?

Answer: B. To see exactly what it changed

Know what changed.

- ✗ A. To rename files: no
- ✗ C. Because diffs fix every bug automatically when they are opened and read by a person: no
- ✗ D. To make the code longer: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-F1-Q5 · question page 1906

AI.CODE.07 · Use commits and diffs with AI changes

A diff shows only "+" lines and no "-" lines. What happened?

Answer: A. Lines were only added

Additions only.

- ✗ B. Nothing changed: lines added
- ✗ C. Every line in the file was deleted and replaced with completely different lines: no removals
- ✗ D. Only comments were removed: no removals

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-F2-Q1 · question page 1907

AI.CODE.07 · Keep history clear and safe

A diff from a tool edit shows changes to a settings file you did not ask about. What should you do?

Answer: D. Review those changes before accepting

Unexpected changes need review.

- ✗ A. Ignore the diff: risky
- ✗ B. Delete the settings file: risky
- ✗ C. Accept everything, since tools only change files that were needed for the requested task: not always

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-F2-Q2 · question page 1908

AI.CODE.07 · Keep history clear and safe

Why prefer small commits?

Answer: C. They are easier to review and undo

One thing per commit.

- ✗ A. Because large commits are not allowed by any version control tool that exists today: not the reason
- ✗ B. Because they hide changes: no
- ✗ D. Because small commits use more storage: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-F2-Q3 · question page 1909

AI.CODE.07 · Keep history clear and safe

A password was committed and then deleted in the next commit. Is it gone?

Answer: A. No; history still has it

Never commit secrets.

- ✗ B. Yes, after one day: no
- ✗ C. Only if the file is small: no
- ✗ D. Yes: history keeps it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-F2-Q4 · question page 1910

AI.CODE.07 · Keep history clear and safe

What should a commit message say?

Answer: B. What changed and why

Clear history.

- ✗ A. Nothing: loses meaning
- ✗ C. Only the programmer's name: recorded already
- ✗ D. Only the date and time: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-F2-Q5 · question page 1911

AI.CODE.07 · Keep history clear and safe

Which history is easiest to understand later?

Answer: D. Small commits with clear messages

Clarity for future readers.

- ✗ A. No commits: no history
- ✗ B. Commits with random messages: hard to follow
- ✗ C. One huge commit called "update": hard to follow

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S92-M-Q1 · question page 1913

AI.CODE.08 · Check that a suggested library exists and is trustworthy

What is a package or dependency?

Answer: B. Code written by others that your program installs and uses

Reusable code from others.

- ✗ A. A box that a computer is shipped in, which contains the cables and the instructions: not the meaning
- ✗ C. An error message: no
- ✗ D. A comment: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

S92-M-Q2 · question page 1914

AI.CODE.08 · Check that a suggested library exists and is trustworthy

What can go wrong when a tool suggests a package?

Answer: A. It may not exist, and attackers can register the name

Invented names can be exploited.

- ✗ B. Nothing, because tools only ever suggest packages that have been checked for safety: not true
- ✗ C. It may have documentation: not a risk
- ✗ D. It may be too popular: not the risk

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

S92-M-Q3 · question page 1915

AI.CODE.08 · Check that a suggested library exists and is trustworthy

What is "slopsquatting"?

Answer: D. Registering package names that AI tools invent

Exploiting hallucinated names.

- ✗ A. Writing very messy code on purpose so that other people find it hard to read and copy: not the meaning
- ✗ B. Sorting a list: no
- ✗ C. Deleting old packages: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

S92-M-Q4 · question page 1916

AI.CODE.08 · Check that a suggested library exists and is trustworthy

Which is a sign of a trustworthy package?

Answer: C. An active project with real documentation and users

Signs of trust.

- ✗ A. A name that looks like a famous one: warning sign
- ✗ B. No description: warning sign
- ✗ D. A brand-new name that nobody has used, with no documentation: warning signs

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

S92-M-Q5 · question page 1917

AI.CODE.08 · Check that a suggested library exists and is trustworthy

Before installing a suggested package, what should you check first?

Answer: D. That it exists and is the intended one

Existence and identity.

- ✗ A. Its logo color: no
- ✗ B. Nothing: must check
- ✗ C. That its name is short enough to type quickly: not the check

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

S92-B1-Q1 · question page 1918

AI.CODE.08 · Compare package checks with checking a contractor

Why check that a contractor exists?

Answer: C. Because you are letting them into your home

Access requires trust.

- ✗ A. Because contractors who exist always do perfect work, so existence is all that matters: not enough
- ✗ B. To learn their age: no
- ✗ D. To get a discount: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

S92-B1-Q2 · question page 1919

AI.CODE.08 · Compare package checks with checking a contractor

A name one letter off from a famous company is like what?

Answer: A. A look-alike package name

Impersonation risk.

- ✗ B. A test: no
- ✗ C. A commit: no
- ✗ D. A well-known package used by thousands of programmers and checked carefully: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

S92-B1-Q3 · question page 1920

AI.CODE.08 · Compare package checks with checking a contractor

Reviews from other customers are like what for a package?

Answer: B. Signs it is used by others

Evidence of trust.

- ✗ A. Its name length: no
- ✗ C. Its error messages: no
- ✗ D. The package's file size: not the comparison

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

S92-B1-Q4 · question page 1921

AI.CODE.08 · Compare package checks with checking a contractor

Installing a package is like what?

Answer: A. Letting someone else's work into your project

Trusting others' code.

- ✗ B. Deleting a file: no
- ✗ C. Running a test: no
- ✗ D. Writing every line of the program yourself: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

S92-B1-Q5 · question page 1922

AI.CODE.08 · Compare package checks with checking a contractor

What is the lesson?

Answer: D. Check a package before installing it

Verify first.

- ✗ A. Install anything suggested: risky
- ✗ B. Only use paid packages: not the point
- ✗ C. Never use code written by anyone else: too strong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

S92-F1-Q1 · question page 1923

AI.CODE.08 · Check suggested packages in practice

A tool suggests "pip install easy-pdf-magic-tools." You find no project page. What should you do?

Answer: B. Do not install it; look for a well-known alternative

No evidence it is real or safe.

- ✗ A. Rename it and install: no
- ✗ C. Install it twice: no
- ✗ D. Install it quickly, since a missing project page just means the package is very new: warning sign

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

S92-F1-Q2 · question page 1924

AI.CODE.08 · Check suggested packages in practice

A suggestion says "reqeusts." What should you notice?

Answer: C. It is a look-alike of "requests"

Misspelled look-alike.

- ✗ A. It is a newer version: not shown
- ✗ B. Nothing, because small spelling differences in package names do not matter at all: they matter
- ✗ D. It is a test file: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

S92-F1-Q3 · question page 1925

AI.CODE.08 · Check suggested packages in practice

Which package choice is safest for a class project?

Answer: D. One the course already uses

Known and trusted.

- ✗ A. A random new package: unknown
- ✗ B. A package that a tool invented a name for, since invented packages are designed for the task: may not exist
- ✗ C. One with no documentation: warning sign

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

S92-F1-Q4 · question page 1926

AI.CODE.08 · Check suggested packages in practice

A tool suggests a package to add two numbers. What is the better choice?

Answer: A. Write the one line yourself

Avoid unnecessary dependencies.

- ✗ B. Skip adding numbers: no
- ✗ C. Install three packages: worse
- ✗ D. Install the package: unneeded risk

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

S92-F1-Q5 · question page 1927

AI.CODE.08 · Check suggested packages in practice

Where should you check that a Python package exists?

Answer: B. Its page on the official package index (PyPI)

Check the official index.

- ✗ A. In the tool's answer itself, since the tool would not suggest a package that did not exist: circular
- ✗ C. Your memory: not reliable
- ✗ D. A random forum: not reliable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

S92-F2-Q1 · question page 1928

AI.CODE.08 · Manage dependency risks over time

A real, popular package has had no updates for years and has open security warnings. What is the concern?

Answer: C. It may be abandoned

Old, unmaintained risk.

- ✗ A. It has too many users: not the concern
- ✗ B. It is too popular: not the concern
- ✗ D. No concern: can be abandoned

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

S92-F2-Q2 · question page 1929

AI.CODE.08 · Manage dependency risks over time

Why record exact package versions?

Answer: D. So builds are repeatable and changes are deliberate

Predictable dependencies.

- ✗ A. To make files longer: no
- ✗ B. To hide which packages are used: no
- ✗ C. Because package managers refuse to install anything: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

S92-F2-Q3 · question page 1930

AI.CODE.08 · Manage dependency risks over time

A student is unsure whether a package is trustworthy. What is the best step?

Answer: B. Ask someone experienced before installing

Ask first.

- ✗ A. Install it and hope: risky
- ✗ C. Share it with classmates: spreads risk
- ✗ D. Install it on the school's shared computer first: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

S92-F2-Q4 · question page 1931

AI.CODE.08 · Manage dependency risks over time

Which name is the most suspicious?

Answer: A. "numpy-official-secure-v2"

Extra words can mark look-alikes.

- ✗ B. "requests": well known
- ✗ C. "numpy," which is a widely used package for working with numbers in Python programs: well known
- ✗ D. "pandas": well known

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

S92-F2-Q5 · question page 1932

AI.CODE.08 · Manage dependency risks over time

Do real packages ever become unsafe?

Answer: C. Yes; they can be compromised

Ongoing vigilance.

- ✗ A. Only if renamed: no
- ✗ B. No; once a package is real and popular, it can never become unsafe in any way at all: they can
- ✗ D. Only on weekends: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

S93-M-Q1 · question page 1934

AI.CODE.09 · Spot exposed secrets and unsafe handling of input

What is an API key?

Answer: C. A code that lets a program use a private or paid service

A credential.

- ✗ A. An error message: no
- ✗ B. A variable name: no
- ✗ D. A keyboard key that programmers press to run their code faster: not the meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-M-Q2 · question page 1935

AI.CODE.09 · Spot exposed secrets and unsafe handling of input

In this example, what is the security problem?

Prepared code example (Python 3) with a fake key. Shown only; nothing runs.

```
API_KEY = "sk-live-4f9a2c7e1b"
print("Connecting with key", API_KEY)
```

Answer: D. The key is written directly in the code

Secrets should be stored separately.

- ✗ A. The variable is called API_KEY in capital letters, which Python does not allow: allowed
- ✗ B. Nothing; keys belong in code: they do not
- ✗ C. The code prints a message: not the main problem

Arithmetic checked by code: not applicable. Code example run and its output checked: yes.
Sources: none (stable)

S93-M-Q3 · question page 1936

AI.CODE.09 · Spot exposed secrets and unsafe handling of input

Where should secrets be stored?

Answer: B. Separately, such as in an environment variable

Outside shared code.

- ✗ A. In a comment at the top of the code, where every programmer can find them quickly: still exposed
- ✗ C. In a chat with an AI tool: never
- ✗ D. In the file name: exposed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-M-Q4 · question page 1937

AI.CODE.09 · Spot exposed secrets and unsafe handling of input

In this example, why is the query unsafe?

Prepared code fragment (Python 3). Shown only; nothing runs.

```
name = input("Name: ")
query = "SELECT * FROM students WHERE name = '" + name + "'"
```

Answer: A. Typed text goes straight into the query

Attackers can change the query.

- ✗ B. The table is called students: irrelevant
- ✗ C. It uses the input function: not by itself
- ✗ D. The query is written in capital letters: not the risk

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-M-Q5 · question page 1938

AI.CODE.09 · Spot exposed secrets and unsafe handling of input

Should you paste a real API key into an AI chat to get help?

Answer: D. No; never paste secrets into a chat

Keep secrets out.

- ✗ A. Yes, if it is short: no
- ✗ B. Only on weekends: no
- ✗ C. Yes, because chats with AI tools are completely private and are never stored anywhere: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-B1-Q1 · question page 1939

AI.CODE.09 · Compare secrets in code with a house key under the doormat

Why is a key under the doormat risky?

Answer: C. Anyone who looks can find it

Easy to find.

- ✗ A. Because keys are heavy: no
- ✗ B. Because doormats damage keys over time, making them stop working in the lock: not the reason
- ✗ D. Because doors are locked: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-B1-Q2 · question page 1940

AI.CODE.09 · Compare secrets in code with a house key under the doormat

A secret written in code is like what?

Answer: B. A key under the doormat

Exposed to anyone who looks.

- ✗ A. A locked safe: no
- ✗ C. A doorbell: no
- ✗ D. A key kept safely: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-B1-Q3 · question page 1941

AI.CODE.09 · Compare secrets in code with a house key under the doormat

Keeping the key in your pocket is like what?

Answer: A. Storing secrets outside the code

Separate storage.

- ✗ B. Giving it away: no
- ✗ C. Posting it online: no
- ✗ D. Writing the secret into a comment: still exposed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-B1-Q4 · question page 1942

AI.CODE.09 · Compare secrets in code with a house key under the doormat

If the key may have been copied, what should you do?

Answer: C. Change the lock

Like rotating a secret.

- ✗ A. Ignore it: risky
- ✗ B. Paint the door: no
- ✗ D. Move the key: still copied

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-B1-Q5 · question page 1943

AI.CODE.09 · Compare secrets in code with a house key under the doormat

In code, "changing the lock" means what?

Answer: A. Replacing the exposed secret with a new one

Rotate the key.

- ✗ B. Adding a comment: no
- ✗ C. Renaming the variable: not enough
- ✗ D. Deleting the line with the secret from the newest version of the code only: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-F1-Q1 · question page 1944

AI.CODE.09 · Handle secrets and input safely in practice

Mia shares her project online and realizes her real API key is in the code. What should she do first?

Answer: D. Change the key at the service right away

Rotate exposed secrets.

- ✗ A. Ignore it: risky
- ✗ B. Delete the key from the code and assume the problem is solved, since nobody saw it yet: may be copied
- ✗ C. Make the repository bigger: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-F1-Q2 · question page 1945

AI.CODE.09 · Handle secrets and input safely in practice

A tool asks you to paste your password so it can "test the login code." What should you do?

Answer: B. Refuse; use a fake test password

Use fake test values.

- ✗ A. Paste it, because tools need the real password to check whether the login code works: never needed
- ✗ C. Paste half of it: still risky
- ✗ D. Email it instead: still risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-F1-Q3 · question page 1946

AI.CODE.09 · Handle secrets and input safely in practice

Which line handles a secret more safely?

Answer: C. Reading the key from an environment variable

Kept outside the code.

- ✗ A. Printing the key at startup: exposed
- ✗ B. Saving the key in the file name: exposed
- ✗ D. Writing the key in a comment next to the code: exposed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-F1-Q4 · question page 1947

AI.CODE.09 · Handle secrets and input safely in practice

A form's code builds a database query by adding typed text. What is the risk?

Answer: D. An attacker can change the query

Unsafe input.

- ✗ A. Nothing: it is a risk
- ✗ B. The text will be misspelled: not the risk
- ✗ C. The query will run more slowly: not the risk

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-F1-Q5 · question page 1948

AI.CODE.09 · Handle secrets and input safely in practice

Generated code contains "API_KEY = 'your-key-here'." What is the safe next step?

Answer: A. Load the real key from a private setting

Keep real keys out of code.

- ✗ B. Delete the line: code may fail
- ✗ C. Share the placeholder: not the issue
- ✗ D. Replace the placeholder text with your real key: exposes it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-F2-Q1 · question page 1949

AI.CODE.09 · Respond to exposed secrets and untrusted input

Why is deleting an exposed key from the code not enough?

Answer: B. It may already have been copied

Rotate the key.

- ✗ A. Because keys grow back: no
- ✗ C. Because code cannot be edited: no
- ✗ D. Because deleting lines of code: possible

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-F2-Q2 · question page 1950

AI.CODE.09 · Respond to exposed secrets and untrusted input

What does "rotate a secret" mean?

Answer: D. Replace it with a new one at the service

Issue a new secret.

- ✗ A. Write it backward: no
- ✗ B. Encrypt the file name: no
- ✗ C. Move the secret to a different line: still exposed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-F2-Q3 · question page 1951

AI.CODE.09 · Respond to exposed secrets and untrusted input

What do parameterized queries do?

Answer: A. Keep user text separate from the command

Safe input handling.

- ✗ B. Print the query: no
- ✗ C. Remove the database: no
- ✗ D. Make the database query run on a schedule every hour without any user input at all: not the meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-F2-Q4 · question page 1952

AI.CODE.09 · Respond to exposed secrets and untrusted input

How should code treat what a user types?

Answer: B. As untrusted

Untrusted input.

- ✗ A. As a secret: no
- ✗ C. As a comment: no
- ✗ D. As fully trusted: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-F2-Q5 · question page 1953

AI.CODE.09 · Respond to exposed secrets and untrusted input

A tutorial's generated code prints the API key at startup "for debugging." What is the risk?

Answer: C. The key can appear in logs or screenshots

Do not print secrets.

- ✗ A. The program runs slower: minor
- ✗ B. No risk, since printed text disappears from the screen as soon as the program finishes: logs keep it
- ✗ D. The key gets longer: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-M-Q1 · question page 1955

AI.CODE.10 · Keep generated code understandable, documented and owned

Is code read or written more often?

Answer: A. Read and changed more often

Maintenance dominates.

- ✗ B. Neither: it is read
- ✗ C. Written more often, because each line of code is written many times and read only once: reversed
- ✗ D. Only written: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-M-Q2 · question page 1956

AI.CODE.10 · Keep generated code understandable, documented and owned

In this example, which function is easier to maintain?

Prepared code example (Python 3). Shown only; nothing runs.

```
def f(a, b):
    return a * b * 1.0825

def price_with_tax(subtotal, quantity):
    TAX_RATE = 0.0825  # local sales tax
    return subtotal * quantity * (1 + TAX_RATE)

print(round(f(10, 2), 2), round(price_with_tax(10, 2), 2))
```

Answer: B. price_with_tax

Clear names and an explained rate.

- ✗ A. Neither: one is clearer
- ✗ C. f, because short names like f, a and b are faster to type and always easier to understand: unclear
- ✗ D. Both equally: one is clearer

Arithmetic checked by code: not applicable. Code example run and its output checked: yes.
Sources: none (stable)

S94-M-Q3 · question page 1957

AI.CODE.10 · Keep generated code understandable, documented and owned

In the same example, do both functions give the same result?

Prepared code example (Python 3). Shown only; nothing runs.

```
def f(a, b):
    return a * b * 1.0825

def price_with_tax(subtotal, quantity):
    TAX_RATE = 0.0825  # local sales tax
    return subtotal * quantity * (1 + TAX_RATE)

print(round(f(10, 2), 2), round(price_with_tax(10, 2), 2))
```

Answer: D. Yes, 21.65

Same math, different clarity.

- ✗ A. No; f gives 20: it gives 21.65
- ✗ B. No; one crashes: both run
- ✗ C. No, because functions with different names always calculate different results: same math

Arithmetic checked by code: yes. Code example run and its output checked: yes. Sources: none (stable)

S94-M-Q4 · question page 1958

AI.CODE.10 · Keep generated code understandable, documented and owned

What is a README?

Answer: C. A document explaining what a project does and how to run it

Project documentation.

- ✗ A. A password file: no
- ✗ B. An error log: no
- ✗ D. A file that the computer reads before every program: not the meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-M-Q5 · question page 1959

AI.CODE.10 · Keep generated code understandable, documented and owned

Who should own a piece of code in use?

Answer: D. A person who understands it and fixes it

Clear responsibility.

- ✗ A. Nobody: needs an owner
- ✗ B. The AI tool that wrote it, which will maintain and fix it forever without any help: tools are not owners
- ✗ C. Whoever used it last: not responsible

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-B1-Q1 · question page 1960

AI.CODE.10 · Compare maintainable code with a well-labeled kitchen

Labeled jars are like what in code?

Answer: A. Clear names and comments

Labels help others.

- ✗ B. Error messages: no
- ✗ C. Passwords: no
- ✗ D. Very short names like x and y: unclear

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-B1-Q2 · question page 1961

AI.CODE.10 · Compare maintainable code with a well-labeled kitchen

Unlabeled jars cause what problem?

Answer: B. Only one person knows what is what

Hidden knowledge.

- ✗ A. Food is cheaper: no
- ✗ C. Everyone in the kitchen always knows exactly what is in every jar without looking: opposite
- ✗ D. Jars break: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-B1-Q3 · question page 1962

AI.CODE.10 · Compare maintainable code with a well-labeled kitchen

The note on the fridge is like what?

Answer: C. A README

How things work.

- ✗ A. A commit message that explains a single change made to one line of the code: not the comparison
- ✗ B. A bug: no
- ✗ D. A test: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-B1-Q4 · question page 1963

AI.CODE.10 · Compare maintainable code with a well-labeled kitchen

The person who keeps the kitchen organized is like what?

Answer: B. The code's owner

Responsibility.

- ✗ A. The fridge: no
- ✗ C. A guest: no
- ✗ D. The AI tool: not the owner

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-B1-Q5 · question page 1964

AI.CODE.10 · Compare maintainable code with a well-labeled kitchen

Salt in the cake is like what in code?

Answer: C. A mistake caused by unclear code

Confusion leads to errors.

- ✗ A. A successful new feature that customers asked for and that works exactly as planned: opposite
- ✗ B. A faster program: no
- ✗ D. A new commit: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-F1-Q1 · question page 1965

AI.CODE.10 · Keep generated code maintainable in practice

A club uses a generated script to track dues, but nobody can explain how it works. What is the risk?

Answer: A. No one can fix it when it breaks

Unowned code is fragile.

- ✗ B. It uses too little memory: not the risk
- ✗ C. No risk, because generated scripts never break and never need to be changed by anyone: they can
- ✗ D. It will run too fast: not the risk

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-F1-Q2 · question page 1966

AI.CODE.10 · Keep generated code maintainable in practice

Which variable name is clearest?

Answer: D. total_price

Says what it holds.

- ✗ A. thing: unclear
- ✗ B. tp: unclear
- ✗ C. x2: unclear

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-F1-Q3 · question page 1967

AI.CODE.10 · Keep generated code maintainable in practice

A generated function does five different jobs. What improves it?

Answer: C. Splitting it into smaller functions

One job per function.

- ✗ A. Renaming it to "do_all": not a fix
- ✗ B. Removing all comments: worse
- ✗ D. Adding a sixth job to the same function: worse

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-F1-Q4 · question page 1968

AI.CODE.10 · Keep generated code maintainable in practice

Which comment is most useful?

Answer: D. "# Round down because the store never charges partial cents"

Explains why.

- ✗ A. "# This line of code does something," which reminds readers that the line is important: uninformative
- ✗ B. "# code": uninformative
- ✗ C. "# hi": uninformative

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-F1-Q5 · question page 1969

AI.CODE.10 · Keep generated code maintainable in practice

Before relying on generated code at work, what should be true?

Answer: B. Someone can explain how it works

Understanding first.

- ✗ A. It should have no comments: no
- ✗ C. It should be secret: no
- ✗ D. It should be long: not a sign

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-F2-Q1 · question page 1970

AI.CODE.10 · Prevent quiet maintenance problems

Code contains 0.0825 with no explanation. What is this called?

Answer: A. A magic number

Unexplained value.

- ✗ B. A test: no
- ✗ C. A comment: no
- ✗ D. A secret key: not a key

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-F2-Q2 · question page 1971

AI.CODE.10 · Prevent quiet maintenance problems

The same calculation is copied in three places. What problem follows?

Answer: B. A fix in one place may be missed in others

Duplication risk.

- ✗ A. It is easier to read: not usually
- ✗ C. The program runs faster: not the point
- ✗ D. Nothing, because copied code always gets updated in every place at the same time: it does not

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-F2-Q3 · question page 1972

AI.CODE.10 · Prevent quiet maintenance problems

A tool generates a 300-line solution for a simple task. What should you ask for?

Answer: A. A simpler version

Simpler is easier to own.

- ✗ B. An even longer version with more features, so it is ready for any future need: harder to maintain
- ✗ C. Nothing; use it: hard to maintain
- ✗ D. A version without tests: worse

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-F2-Q4 · question page 1973

AI.CODE.10 · Prevent quiet maintenance problems

The student who built the class website is graduating. What should happen?

Answer: C. Hand over knowledge, not just files

Transfer understanding.

- ✗ A. Change the password and tell no one: loses access
- ✗ B. Delete the website: unnecessary
- ✗ D. Nothing: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-F2-Q5 · question page 1974

AI.CODE.10 · Prevent quiet maintenance problems

Why add tests to generated code you will keep?

Answer: D. They describe and protect how it should behave

Tests document behavior.

- ✗ A. To slow it down: no
- ✗ B. To hide bugs: no
- ✗ C. Because tests make the code shorter and remove the need for any comments or README: not the point

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-M-Q1 · question page 1976

AI.AGENT.01 · Tell a chatbot, a fixed workflow and an agent apart

What does a chatbot do?

Answer: C. Answers in a conversation while you decide what to do

You act on its replies.

- ✗ A. Prints documents: no
- ✗ B. Follows a fixed schedule: that is a workflow
- ✗ D. Takes actions in the world on its own, such as sending emails: that is an agent

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-M-Q2 · question page 1977

AI.AGENT.01 · Tell a chatbot, a fixed workflow and an agent apart

What makes a workflow "fixed"?

Answer: D. A person designed its steps in advance

Predetermined steps.

- ✗ A. It never uses a model: it may
- ✗ B. It has no steps: it has steps
- ✗ C. The model chooses each new step by itself: that is an agent

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-M-Q3 · question page 1978

AI.AGENT.01 · Tell a chatbot, a fixed workflow and an agent apart

What makes something an agent?

Answer: B. It chooses its own next steps toward a goal

Goal plus self-chosen steps.

- ✗ A. It has a friendly name: no
- ✗ C. It runs on a timer: that is scheduling
- ✗ D. It answers one question at a time and waits: that is a chatbot

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-M-Q4 · question page 1979

AI.AGENT.01 · Tell a chatbot, a fixed workflow and an agent apart

"When a form arrives, summarize it, then email the manager" describes what?

Answer: A. A tool-using workflow

Fixed steps with a model inside.

- ✗ B. A chatbot: no conversation
- ✗ C. A search engine: no
- ✗ D. An agent: steps are fixed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-M-Q5 · question page 1980

AI.AGENT.01 · Tell a chatbot, a fixed workflow and an agent apart

Which needs the most limits and oversight?

Answer: D. An agent

More freedom, more risk.

- ✗ A. A calculator: no
- ✗ B. A fixed workflow: more predictable
- ✗ C. A chatbot: fewer actions

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-B1-Q1 · question page 1981

AI.AGENT.01 · Compare the three with a GPS, a delivery route and a personal assistant

A GPS you ask for directions is like what?

Answer: B. A chatbot

It suggests; you act.

- ✗ A. A calendar: no
- ✗ C. A workflow: not fixed stops
- ✗ D. An agent: you drive

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-B1-Q2 · question page 1982

AI.AGENT.01 · Compare the three with a GPS, a delivery route and a personal assistant

A fixed delivery route is like what?

Answer: C. A workflow

Designed in advance.

- ✗ A. A map: not the comparison
- ✗ B. A chatbot: no conversation
- ✗ D. An agent: fixed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-B1-Q3 · question page 1983

AI.AGENT.01 · Compare the three with a GPS, a delivery route and a personal assistant

An assistant choosing taxis and trains is like what?

Answer: A. An agent

Goal plus chosen steps.

- ✗ B. A timetable: no
- ✗ C. A fixed workflow: not fixed
- ✗ D. A chatbot: it chooses

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-B1-Q4 · question page 1984

AI.AGENT.01 · Compare the three with a GPS, a delivery route and a personal assistant

Which option is most flexible?

Answer: B. The personal assistant

Chooses its own steps.

- ✗ A. All are equal: they differ
- ✗ C. The GPS: suggests only
- ✗ D. The fixed delivery route: fixed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-B1-Q5 · question page 1985

AI.AGENT.01 · Compare the three with a GPS, a delivery route and a personal assistant

Whose choices most need checking?

Answer: A. The assistant's

Most freedom, most checking.

- ✗ B. The route's: predictable
- ✗ C. No one's: checking matters
- ✗ D. The GPS's: you decide

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-F1-Q1 · question page 1986

AI.AGENT.01 · Classify AI setups in practice

Lena types questions to a study helper and decides what to write. What is it?

Answer: C. A chatbot

Conversation; she acts.

- ✗ A. A fixed workflow: no fixed steps
- ✗ B. A spreadsheet: no
- ✗ D. An agent: she decides

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-F1-Q2 · question page 1987

AI.AGENT.01 · Classify AI setups in practice

Every night, a system collects new orders, uses a model to tag them, and files them. What is it?

Answer: D. A fixed workflow

Same steps each time.

- ✗ A. A chatbot: no conversation
- ✗ B. A calculator: no
- ✗ C. An agent: steps are fixed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-F1-Q3 · question page 1988

AI.AGENT.01 · Classify AI setups in practice

Told "book the cheapest flight to Denver next week," a system decides which searches to run, changes its plan when a flight is sold out, and then books. What is it?

Answer: C. An agent

It is given a goal and chooses and adjusts its own steps.

- ✗ A. A search engine: it also books
- ✗ B. A fixed workflow: it chooses steps
- ✗ D. A chatbot: it acts

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-F1-Q4 · question page 1989

AI.AGENT.01 · Classify AI setups in practice

Which system can affect the world most directly?

Answer: D. An agent that can send and buy

Actions in the world.

- ✗ A. A printed manual: no
- ✗ B. A calculator: no
- ✗ C. A chatbot that only shows text answers on a screen for the person to read and decide: text only

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-F1-Q5 · question page 1990

AI.AGENT.01 · Classify AI setups in practice

A workflow uses a model only to summarize; a person designed every other step. What decides the order of steps?

Answer: A. The person's design

Fixed design.

- ✗ B. Chance: no
- ✗ C. The customer: no
- ✗ D. The model, which rearranges the order of the steps every time the workflow is started: fixed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-F2-Q1 · question page 1991

AI.AGENT.01 · Look past labels to what a system can do

A product is marketed as an "agent" but always runs the same three steps. What is it really?

Answer: B. A fixed workflow

Look at behavior.

- ✗ A. A database: no
- ✗ C. A chatbot: no conversation
- ✗ D. An agent: labels are loose

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-F2-Q2 · question page 1992

AI.AGENT.01 · Look past labels to what a system can do

Which question best reveals the risk of a system?

Answer: D. "Can it act without a person approving?"

Unapproved actions matter.

- ✗ A. "Is it new?": not the risk
- ✗ B. "Is it popular?": not the risk
- ✗ C. "Does it have a friendly name and a nice logo?": irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-F2-Q3 · question page 1993

AI.AGENT.01 · Look past labels to what a system can do

A chatbot quietly searches the web before answering. Does that make it an agent?

Answer: A. Not necessarily; it depends on who chooses the next steps

An agent chooses its own steps toward a goal; that can be true even when it only returns text.

- ✗ B. No, chatbots cannot use tools: they can
- ✗ C. Yes, because any system that uses a tool is an agent: using a tool does not decide it
- ✗ D. Yes, if it is fast: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-F2-Q4 · question page 1994

AI.AGENT.01 · Look past labels to what a system can do

Which ability needs the most limits?

Answer: B. Deleting files

Irreversible actions.

- ✗ A. Answering a quiz question: low risk
- ✗ C. Suggesting a title: low risk
- ✗ D. Showing a short text summary: low risk

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-F2-Q5 · question page 1995

AI.AGENT.01 · Look past labels to what a system can do

Who decides the next step tells you what?

Answer: C. How much checking the system needs

Decision-maker shapes oversight.

- ✗ A. Nothing important, since every AI system needs exactly the same amount of checking: it varies
- ✗ B. Its speed: no
- ✗ D. Its price: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-M-Q1 · question page 1997

AI.AGENT.02 · Describe an agent's goal, tools, plan, feedback and stopping condition

"Find three venues under 500 dollars for 40 people" is which part?

Answer: B. Goal

What it is trying to achieve.

- ✗ A. Tool, because finding venues is something that a search tool does for the agent: it is the aim
- ✗ C. Feedback: no
- ✗ D. Stopping condition: partly, but it is the goal

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-M-Q2 · question page 1998

AI.AGENT.02 · Describe an agent's goal, tools, plan, feedback and stopping condition

Search, calendar and email are which part?

Answer: D. Tools

What it can use.

- ✗ A. Plan: no
- ✗ B. Feedback: no
- ✗ C. Goals: they are means

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-M-Q3 · question page 1999

AI.AGENT.02 · Describe an agent's goal, tools, plan, feedback and stopping condition

"A venue is full, so look again" is which part?

Answer: C. Feedback

Results guide next steps.

- ✗ A. Goal, because looking again for another venue is the main thing the agent wants to do: it is a response
- ✗ B. Tool: no
- ✗ D. Stopping condition: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-M-Q4 · question page 2000

AI.AGENT.02 · Describe an agent's goal, tools, plan, feedback and stopping condition

"Stop after 20 searches" is which part?

Answer: A. Stopping condition

When to stop.

- ✗ B. Goal: no
- ✗ C. Tool: no
- ✗ D. Plan, because twenty searches is the list of steps the agent chooses to take in order: it is a limit

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-M-Q5 · question page 2001

AI.AGENT.02 · Describe an agent's goal, tools, plan, feedback and stopping condition

What can a missing stopping condition cause?

Answer: C. Wandering or repeating steps

No clear end.

- ✗ A. Fewer tools: no
- ✗ B. A shorter plan: no
- ✗ D. Faster and better results: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-B1-Q1 · question page 2002

AI.AGENT.02 · Compare an agent's parts with sending someone grocery shopping

"Ingredients for tacos for 6" is which part?

Answer: B. Goal

The aim.

- ✗ A. Feedback: no
- ✗ C. Tool: no
- ✗ D. Plan: it is the aim

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-B1-Q2 · question page 2003

AI.AGENT.02 · Compare an agent's parts with sending someone grocery shopping

Money and a car are which part?

Answer: D. Tools

What can be used.

- ✗ A. Stopping condition: no
- ✗ B. Goals, because the friend is trying to get money and a car during the trip: they are means
- ✗ C. Feedback: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-B1-Q3 · question page 2004

AI.AGENT.02 · Compare an agent's parts with sending someone grocery shopping

"Out of tortillas, so try the bakery" is which part?

Answer: A. Feedback

Results change the plan.

- ✗ B. Stopping condition, since running out of tortillas means the shopping is finished: it continues
- ✗ C. Goal: no
- ✗ D. Tool: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-B1-Q4 · question page 2005

AI.AGENT.02 · Compare an agent's parts with sending someone grocery shopping

"Come back after spending 40 dollars" is which part?

Answer: C. Stopping condition

A budget limit.

- ✗ A. Tool: no
- ✗ B. Feedback: no
- ✗ D. Goal, because spending forty dollars is the main reason for going shopping at all: it is a limit

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-B1-Q5 · question page 2006

AI.AGENT.02 · Compare an agent's parts with sending someone grocery shopping

What happens without a stopping condition?

Answer: A. Your friend may keep searching too long

No clear end.

- ✗ B. The tacos taste better: no
- ✗ C. Nothing changes: it matters
- ✗ D. Your friend will always come back sooner: not likely

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-F1-Q1 · question page 2007

AI.AGENT.02 · Describe and improve agent setups in practice

An agent is told "organize my inbox" with no other details. What is missing?

Answer: B. A clear goal and stopping condition

Vague goals wander.

- ✗ A. More tools, because an agent with access to more tools always understands vague goals: not the fix
- ✗ C. Nothing: it is vague
- ✗ D. A shorter name: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-F1-Q2 · question page 2008

AI.AGENT.02 · Describe and improve agent setups in practice

Which stopping condition is clearest?

Answer: D. "Stop after sorting 200 emails or 10 minutes."

Measurable limits.

- ✗ A. "Stop sometime.": vague
- ✗ B. "Never stop.": no limit
- ✗ C. "Stop whenever you feel that the inbox looks nice.": vague

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-F1-Q3 · question page 2009

AI.AGENT.02 · Describe and improve agent setups in practice

An agent searching for flights finds one sold out and searches again. What part is at work?

Answer: A. Feedback

Results guide next steps.

- ✗ B. Stopping condition: no
- ✗ C. Goal, because a sold-out flight is a new goal that the agent must achieve next: it is a result
- ✗ D. Tool: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-F1-Q4 · question page 2010

AI.AGENT.02 · Describe and improve agent setups in practice

An agent can use a calendar but not email. Which part does this describe?

Answer: C. Tools

What it can use.

- ✗ A. Goal: no
- ✗ B. Feedback: no
- ✗ D. Plan: it is access

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-F1-Q5 · question page 2011

AI.AGENT.02 · Describe and improve agent setups in practice

Before a high-stakes task, what is a sensible request?

Answer: B. "Show me your plan before acting."

Review the plan first.

- ✗ A. "Act as fast as possible.": no review
- ✗ C. "Use every tool you can.": risky
- ✗ D. "Go ahead with whatever plan you think is best, and tell me about it next week.": no review

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-F2-Q1 · question page 2012

AI.AGENT.02 · Prevent misread goals and runaway loops

An agent told "get the most sign-ups" sends ten emails to each person. What went wrong?

Answer: D. The goal had no limits

Add limits to goals.

- ✗ A. The sign-up form was wrong: not stated
- ✗ B. The emails were too short: not the issue
- ✗ C. The agent was broken: it followed the goal

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-F2-Q2 · question page 2013

AI.AGENT.02 · Prevent misread goals and runaway loops

Which goal includes a useful limit?

Answer: C. "Invite club members, no more than one email each."

Bounded goal.

- ✗ A. "Invite as many people as you possibly can, using any method you think will work.": no limit
- ✗ B. "Do your best.": vague
- ✗ D. "Invite people.": vague

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-F2-Q3 · question page 2014

AI.AGENT.02 · Prevent misread goals and runaway loops

What should a stopping condition include?

Answer: A. A budget of time, steps or money

Hard limits.

- ✗ B. Only a friendly message: not a limit
- ✗ C. Nothing specific, since agents always know by themselves when it is the right time to stop: not reliable
- ✗ D. A longer goal: not a limit

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-F2-Q4 · question page 2015

AI.AGENT.02 · Prevent misread goals and runaway loops

An agent keeps repeating the same search without progress. What would have prevented this?

Answer: D. A step limit in its stopping condition

Stop instead of loop.

- ✗ A. A faster computer: faster loops
- ✗ B. A longer goal: not the fix
- ✗ C. Giving the agent more tools to use: not the fix

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)